
First-ish Order Methods: Hessian-aware Scalings of Gradient Descent

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Abstract

Gradient descent is the primary workhorse for optimizing large-scale problems in machine learning. However, its performance is highly sensitive to the choice of the learning rate. A key limitation of gradient descent is its lack of natural scaling, which often necessitates expensive line searches or heuristic tuning to determine an appropriate step size. In this paper, we address this limitation by incorporating Hessian information to scale the gradient direction. By accounting for the curvature of the function along the gradient, our adaptive, Hessian-aware scaling method ensures a local unit step size guarantee, even in nonconvex settings. Near a local minimum that satisfies the second-order sufficient conditions, our approach achieves linear convergence with a unit step size. We show that our method converges globally under a significantly weaker version of the standard Lipschitz gradient smoothness assumption. Even when Hessian information is inexact, the local unit step size guarantee and global convergence properties remain valid under mild conditions. Finally, we validate our theoretical results empirically on a range of convex and nonconvex machine learning tasks.

1 Introduction

Consider the optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}), \quad (1)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is twice continuously differentiable, bounded below, and possibly nonconvex. Arguably, the simplest and most widely used workhorse for solving such problems in large-scale machine learning is gradient descent (GD) and its stochastic variants [41]. Recall that an iteration of GD takes the form

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{g}_k,$$

where $\mathbf{g}_k \triangleq \mathbf{g}(\mathbf{x}_k) = \nabla f(\mathbf{x}_k)$ is the gradient of f at $\mathbf{x}_k$ and α_k is a step size. Gradient descent is favored for its simplicity, low per-iteration cost, and convergence properties in nonconvex settings. With an arbitrary initialization, $\mathbf{x}_0$, the only hyperparameter is the step size, which may vary across

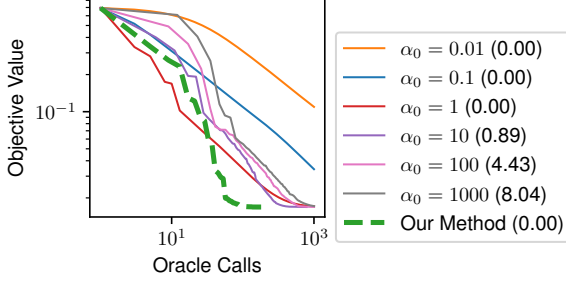


Figure 1: Logistic regression on the Mushrooms dataset [1]. We compare our scaling approach to standard GD with backtracking line search, using different initialization of the backtracking procedure (α^0), and assess performance based on oracle calls (Appendix D.1). The average number of backtracking steps per iteration is reported in brackets.

iterations. Choosing this parameter has been the focus of significant research. Often GD is analyzed using the Lipschitz smoothness of the gradient to select a constant step size that guarantees a monotonic decrease in function value. However, this strategy is typically impractical, as it relies on knowing the Lipschitz constant and results in overly conservative steps [46]. In modern machine learning, learning rates and schedules are usually tuned through trial and error. While this can deliver strong empirical performance, it incurs significant computational cost, lacks theoretical justification, and must be repeated whenever the model or dataset changes.

Backtracking line search is a principled and often more robust alternative [54] to constant step sizes and trial-and-error tuning. It comes with theoretical convergence guarantees, does not require problem-specific constants, and offers some adaptivity to local geometry [24], leading to a recent resurgence of interest in deep learning applications [25, 68]. A key challenge, however, lies in the choice of initial step size in the backtracking procedure. If it is too large, many backtracking steps are needed; if too small, the method may fail to explore larger step sizes, as illustrated in Figure 1. To mitigate this, various initialization strategies have emerged, including heuristics [54, Chapter 3], resetting schemes [68], and Polyak-based methods [25], though the latter requires a knowledge of a lower bound on the objective function as well as an explicit limit on the step size.

In the case of line search applied to second-order methods, the situation is markedly different. These methods incorporate local curvature information to construct properly scaled search directions, for which a unit step size yields a sufficient decrease in the objective function¹. This property makes second-order methods notably less sensitive to hyperparameter choices [72] and allows the unit step to serve as a natural initial guess in backtracking line search [10, 54]. Indeed, practical experience suggests that backtracking is typically only required for a handful of iterations when using Newton’s method. Unfortunately, second-order methods are often impractical for large-scale problems.

Together, these facts lead us to the motivating question for our paper: *Can we employ local curvature to properly, and efficiently, scale the gradient itself?* To that end, we propose our “first-ish order” method, which employs updates of the form:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k, \quad \mathbf{p}_k = -s_k \mathbf{g}_k, \quad (2)$$

where $s_k > 0$ is an adaptive, *Hessian-aware* scaling and $\alpha_k > 0$ is a step size chosen by line search, to ensure global convergence. Remarkably, we show that certain Hessian-aware scalings can endow GD with a local unit step size guarantee in a similar manner to second-order methods. Our approach is extremely simple and only requires access to local, directional curvature information, which can be calculated with a single Hessian-vector product. The key to our results is the fact that the scaled gradient direction, $\mathbf{p}_k$, satisfies the following *second-order descent* condition²:

$$\langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle \leq 0, \quad (3)$$

where $\mathbf{H}_k \triangleq \mathbf{H}(\mathbf{x}_k) = \nabla^2 f(\mathbf{x}_k)$ is the Hessian of f at $\mathbf{x}_k$. When curvature along $\mathbf{p}$ is positive, (3) strengthens the standard first-order descent condition $\langle \mathbf{g}_k, \mathbf{p}_k \rangle < 0$. The condition (3) is satisfied by search directions in second-order methods such as Newton’s method and its inexact variants (e.g., Newton-MR, Newton-CG) when non-positive curvature is not encountered [43, 54]. It avoids crude

¹This brings to mind the following quote about the significance of a unit step size: “Any optimization algorithm for which the unit step length works has some wisdom. It is too much of a fluke if the unit step length [accidentally] works.” - J. Nocedal (Source.)

²Note that (3) has also been called “2.5th order descent” as it is a stronger condition than the typical reduction in the local quadratic approximation to the objective.

curvature bounds and enables second-order analysis without requiring problem-specific constants. Our method handles negative curvature by direct verification along the gradient direction, allowing for use in nonconvex settings without intrusive modification of the Hessian.

Contributions. Our contributions are outlined as follows:

1. In Section 3.1, we show that when the gradient is small, our scaled gradient direction satisfies the Armijo condition with unit step size, even for nonconvex f . We also prove global convergence under generalized smoothness assumptions that extend beyond standard Lipschitz continuity of the gradient and Hessian. An extension to inexact Hessians is discussed in Appendix B.7.
2. In Section 3.2, we establish local convergence results for our method. Specifically, we show that under certain conditions near a minimum, the unit step size is consistently accepted by the line search condition, leading to local linear convergence of the function value. Furthermore, we show our method can potentially attain an improved rate over standard gradient descent. For a particular scaling, we also prove local linear convergence of the gradient norm.
3. Finally, in Section 4, we validate our results numerically on large scale, nonconvex objectives arising from problems in data science.

2 Hessian-aware Scaling

We now introduce and motivate our Hessian-aware scalings and highlight their key properties.

2.1 Our Approach

To determine the scaling, s_k , in (2) required for the update direction, we consider solving the Newton system subproblem with the search direction fixed to the negative gradient. We also account for varying curvature along the gradient direction, by considering the cases of strongly positive, limited positive, and negative curvature separately.

Strong Positive Curvature (SPC). For strongly convex problems, the Newton direction minimizes the local quadratic approximation: $\min_{\mathbf{p}} \langle \mathbf{p}, \mathbf{g} \rangle + \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle / 2$. Since our step direction is fixed to the negative gradient, we restrict $\mathbf{p} = -s\mathbf{g}$, and, assuming $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > 0$, optimize over s to obtain

$$s^{\text{CG}} = \frac{\|\mathbf{g}\|^2}{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle}. \quad (4)$$

The notation s^{CG} reflects a connection to the conjugate gradient (CG) method. Indeed, the CG scaled direction, $\mathbf{p} = -s^{\text{CG}}\mathbf{g}$ is equivalent to the first inner iteration of Newton-CG, an inexact Newton method based on applying CG to the Newton system [54]. Alternatively, we can also view the Newton direction as the solution to the least squares problem $\min_{\mathbf{p}} \|\mathbf{H}\mathbf{p} + \mathbf{g}\|^2$. Applying the restriction, $\mathbf{p} = -s\mathbf{g}$, and optimizing over s yields an alternative scaling

$$s^{\text{MR}} = \frac{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle}{\|\mathbf{H}\mathbf{g}\|^2}. \quad (5)$$

The notation s^{MR} highlights a connection to the minimum residual (MINRES) method [57]. In fact, $\mathbf{p} = -s^{\text{MR}}\mathbf{g}$ is the search direction produced on the first inner iteration of Newton-MR, an inexact Newton method which employs MINRES as a subproblem solver [42, 43, 63, 66]. The link between the CG and MR scalings and inexact second-order methods is an impetus for our study of scaled gradient methods. Indeed, our curvature validation was inspired by the treatment of limited positive curvature in [42, 66].

From these two fundamental scalings, additional factors like the geometric mean of CG and MR scalings can be derived:

$$s^{\text{GM}} = \sqrt{s^{\text{MR}} s^{\text{CG}}} = \frac{\|\mathbf{g}\|}{\|\mathbf{H}\mathbf{g}\|}. \quad (6)$$

The CG, GM, and MR scalings each reflect an aspect of the inverse Hessian restricted to one dimension. For example, the CG scaling in (4) is an inverse Rayleigh quotient of the Hessian, $\mathbf{H}$,

with respect to $\mathbf{g}$, while the MR scaling in (5) corresponds to a Rayleigh quotient of the Hessian *pseudo-inverse*, $\mathbf{H}^\dagger$, with respect to $\mathbf{H}\mathbf{g}$ (using $\mathbf{H}\mathbf{H}^\dagger\mathbf{H} = \mathbf{H}$). In the univariate case the update in (2) collapses to the exact Newton step for all scalings. Thus, these scalings are most effective under SPC along the gradient direction, i.e., $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > \sigma \|\mathbf{g}\|^2$ for some $\sigma > 0$. The SPC condition can be interpreted as a strong convexity condition *restricted to the gradient direction*.

Negative Curvature (NC). In nonconvex settings, the indefiniteness of the Hessian can make the gradient a NC direction, i.e., $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle < 0$. Since $-\mathbf{g}$ is already a descent direction, both the first and second directional derivatives along $-\mathbf{g}$ are negative, regardless of scaling choice. Previous theoretical results [15, 28, 43] and practical experience suggest that substantial progress can be made with large steps along negative curvature directions.

Limited Positive Curvature (LPC). The LPC case, where $0 \leq \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq \sigma \|\mathbf{g}\|^2$, represents a middle ground between SPC and NC, characterized by small but non-negative curvature along $\mathbf{g}$. Here, σ acts like a gradient Lipschitz constant, constraining the second-order term in the Taylor expansion along $-\mathbf{g}$. Since second-order information is unreliable in this regime, we revert to gradient descent with a step size independent of second-order information. Thanks to the curvature bound σ , scalings satisfying $s \leq 1/\sigma$ maintain second-order descent properties akin to those in the SPC and NC cases.

The above discussions are summarized in Algorithm 1. At first glance, Algorithm 1 comes with a number of hyperparameters. However most of these hyperparameters come with natural settings and do not require tuning. We detail these settings in Appendix A.1.

Algorithm 1 Hessian-aware Scaling Selection

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1: Inputs: Gradient  $\mathbf{g}$ , Hessian  $\mathbf{H}$ , and SPC scaling tolerance  $\sigma > 0$ .
2: Set the range for LPC scaling as  $s_{\min}^{\text{LPC}} \in (0, 1/\sigma)$ .
3: Set the range for NC scaling as  $0 < s_{\min}^{\text{NC}} \leq s_{\max}^{\text{NC}} < \infty$ .
4: if  $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > \sigma \|\mathbf{g}\|^2$  then
5:   choose  $s^{\text{SPC}} \in \{s^{\text{MR}}, s^{\text{CG}}, s^{\text{GM}}\}$ , set  $\mathbf{p} = -s^{\text{SPC}}\mathbf{g}$ , set FLAG = SPC.
6: else if  $0 \leq \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq \sigma \|\mathbf{g}\|^2$  then
7:   choose  $s^{\text{LPC}} \in [s_{\min}^{\text{LPC}}, 1/\sigma]$ , set  $\mathbf{p} = -s^{\text{LPC}}\mathbf{g}$ , set FLAG = LPC.
8: else
9:   choose  $s^{\text{NC}} \in [s_{\min}^{\text{NC}}, s_{\max}^{\text{NC}}]$ , set  $\mathbf{p} = -s^{\text{NC}}\mathbf{g}$ , set FLAG = NC.
10: end if
11: Return  $\mathbf{p}$ , FLAG.

```

Remark 2.1. Computationally, Algorithm 1 requires one additional Hessian-vector product, $\mathbf{H}\mathbf{g}$, which can be computed efficiently without forming the full Hessian using automatic differentiation [7, 61]. This involves an extra forward and backward pass through the computational graph and some additional memory overhead. While this adds some cost to each iteration, our experiments (Section 4) show that our scaling method largely eliminates the need for backtracking in line search, reducing the total number of function evaluations (forward passes) compared to unscaled line search.

Basic Properties We now collect some basic properties of the scalings produced by Algorithm 1. We relegate all proofs to Appendix A.

Proposition 2.2 (Scaling Upper Bounds). *In the SPC case, $0 < s^{\text{MR}} \leq s^{\text{GM}} \leq s^{\text{CG}} \leq 1/\sigma$, while for the LPC case, $0 < s^{\text{LPC}} \leq 1/\sigma$. Finally, in the NC case, $0 < s^{\text{NC}} \leq s_{\max}^{\text{NC}}$.*

The proof of this result demonstrates the importance of validating the SPC condition to obtain an upper bound in the SPC and LPC cases.

Proposition 2.3 (Second-order Descent). *Suppose $\mathbf{g} \neq 0$. The direction $\mathbf{p}$ returned by Algorithm 1 satisfies both the first-order descent condition $\langle \mathbf{g}, \mathbf{p} \rangle < 0$ and the second-order descent condition (3).*

Remark 2.4. Our selection of scalings satisfying second-order descent is not exhaustive. For instance, if f has an $L_{\mathbf{g}}$ -Lipschitz gradient, $s = 1/L_{\mathbf{g}}$ ensures (3) by conservatively controlling curvature. However, this choice is not adaptive to local curvature, and $L_{\mathbf{g}}$ may be unknown.

A well-known property of the classical Newton step is its invariance under affine transformations [10, Chapter 9.5]. While directions parallel to gradient cannot achieve full affine invariance for general objectives, Proposition 2.5 shows that our Hessian-aware scalings lead to invariance to *scalar* transformations.

Proposition 2.5 (Scalar Invariance). *Consider (2) with $s_k = s_k^{\text{SPC}}$ for all k , applied to $f(\mathbf{x})$ and its scalar reparameterization $f(\mathbf{y})$ where $\mathbf{y} = \mathbf{x}/c$ for any $c \neq 0$. Then $\mathbf{y}_k = \mathbf{x}_k/c$ for all k .*

As shown in Proposition 2.5, our method is fully invariant to scalar coordinate transformations in the strongly convex setting, where SPC scalings are used at each step. This contrasts with standard gradient descent, which is highly sensitive to such changes.

2.2 Related Works

Quadratic Problems. The scalings (4) and (5) are well-studied for strongly convex quadratic functions; see [26, 45] and references therein. In fact, the scalings correspond to exact line search methods along the gradient direction for f and $\|\mathbf{g}\|^2$, known as steepest descent and minimal gradient methods, respectively³. The GM scaling (6) has also been studied for such problems by [17], showing that for $\alpha > 0$, s^{GM} estimates an inverse gradient Lipschitz constant.

Barzilai-Borwein Methods. The Barzilai-Borwein (BB) method [6, 16, 23] is also based on scalar minimization of a second-order approximation. For quadratics, the long and short BB step sizes are equivalent to our CG and MR scalings, shifted by one iteration, due to their equivalence to steepest descent and minimal gradient methods. While BB achieves R-linear convergence for strongly convex quadratics, its convergence for non-quadratic objectives cannot be guaranteed [11].

Smoothness and Adaptivity. The global convergence analysis in this paper does not rely on the conventional Lipschitz gradient smoothness assumption, which has come under increasing scrutiny in recent works. Many machine learning objectives, such as feedforward or recurrent neural networks, fail to satisfy this condition [60]. Recent studies suggest that this assumption may not hold even along the optimization trajectory [14]. Instead, Ahn et al. [2, Remark 5] argue that *Lipschitz Hessian smoothness* is more appropriate; our work adopts a weaker form of this assumption (Assumption 3.4). In line with these ideas, Patel and Berahas [59] study gradient descent with diminishing step sizes under local Lipschitz smoothness, a weaker assumption than the standard one.

Like our method, a number of works have incorporated local information into step size selection, e.g., the Polyak step size [62] and its variants [44, 55, 56]. Malitsky and Mishchenko [46, 47] adaptively set step sizes using local Lipschitz estimates and control sequences, ensuring global convergence under convexity and locally Lipschitz gradients. Mishkin et al. [50] explores adaptive step sizes via directional smoothness, closely related to our use of gradient curvature. Grimmer [29] show that larger step sizes can improve rates, while Altschuler and Parrilo [3] demonstrate that combining long and short steps enhances convergence in convex settings. Berahas et al. [8] investigates local first-order smoothness oracles, while methods like D-adaptation [20, 49] and Do(W)G [35, 38] achieve parameter-free convergence by adaptively estimating problem constants.

Quadratic Model Scaling Utilizing a quadratic model to rescale search directions is not new. The KFAC method [48] rescales directions using a quadratic model based on the Fisher information matrix. Similarly, Roulet et al. [65] adaptively set the step size using a quadratic model of the objective along a search direction, akin to our CG scaling. Roulet et al. [65] show that curvature-aware scaling can align optimization dynamics with the edge of stability [14], though they do not exploit negative curvature or provide theoretical guarantees. Castera et al. [13] use a quadratic model (similar to our CG scaling) to verify curvature, enabling BB step sizes in nonconvex and stochastic settings, and also find that large step sizes are viable with negative curvature. de Gournay and Gossard [19] rescale step directions using directional second-order information, applying a moving average of directional curvatures, but focus on practical implementation without convergence theory.

³Beyond quadratics, there is no correspondence to exact line search, so we avoid labeling (4) and (5) as such.

3 Convergence Analyses

In Section 3.1, we study the global behavior of (2) with Algorithm 1 and the classical Armijo line search; local convergence is addressed in Section 3.2. Proofs are deferred to Appendix B. We note that an extension to the case where the Hessian is only known inexactly is provided in Appendix B.7.

3.1 Unit Step Size Guarantee and Global Convergence

To globalize the iteration in (2), we select a step size via the Armijo condition, which requires α to satisfy a sufficient reduction in the function value

$$f(\mathbf{x} + \alpha \mathbf{p}) \leq f(\mathbf{x}) + \rho \alpha \langle \mathbf{p}, \mathbf{g} \rangle, \quad (7)$$

for some constant $\rho \in (0, 1/2)$. The resulting method is depicted in Algorithm 2, where backtracking (Algorithm 3 in Appendix B.1) and forward tracking (Algorithm 4 in Appendix B.1) strategies are used to find α_k satisfying (7).

Algorithm 2 Scaled Gradient Descent With Line Search

- 1: **Input:** Line search parameter $\rho < 1/2$, termination tolerance $\varepsilon_g > 0$, initialization $\mathbf{x}_0, k = 0$.
 - 2: **while** $\|\mathbf{g}_k\| \geq \varepsilon_g$ **do**
 - 3: $[\mathbf{p}_k, \text{FLAG}] \leftarrow$ Call Algorithm 1 with $\mathbf{H}_k, \mathbf{g}_k$ and σ
 - 4: For FLAG = SPC/LPC, use Algorithm 3 with $\alpha^0 = 1$ to find $\alpha_k \in (0, 1]$ satisfying (7). For FLAG = NC, use Algorithm 4 with $\alpha^0 = 1$ to find $\alpha_k \in (0, \infty)$ satisfying (7).
 - 5: $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$
 - 6: $k \leftarrow k + 1$
 - 7: **end while**
-

The primary assumption our analysis is built on is a weakened version of the typical Lipschitz Hessian smoothness condition, which requires smoothness to hold only along the negative-gradient direction.

Assumption 3.1 (Hessian Directional Smoothness). There exists, $0 \leq L_2 < \infty$ such that, $\forall \mathbf{x} \in \mathbb{R}^d$ and $\forall t \geq 0$ we have $\|\mathbf{H}(\mathbf{x} - t\mathbf{g}(\mathbf{x})) - \mathbf{H}(\mathbf{x})\| \leq tL_2 \|\mathbf{g}(\mathbf{x})\|$.

Remark 3.2. While Assumption 3.1 is difficult to verify directly, it is implied by L_H -Hessian Lipschitz continuity. Indeed, $L_2 \leq L_H$ since Assumption 3.1 applies along a single direction at each $\mathbf{x}$.

Our main result states that, under only Assumption 3.1, the unit step size, $\alpha = 1$, locally ensures sufficient decrease in the function value for the scaled gradient direction, $\mathbf{p}$, produced by Algorithm 1.

Theorem 3.3 (Sufficient Condition for Acceptance of Unit Step Size). *Consider Assumption 3.1 and let $\mathbf{p}$ be the direction selected by Algorithm 1. The Armijo condition (7) is satisfied with $\alpha = 1$ if*

$$\|\mathbf{g}\| \leq \min \left\{ \frac{6\sigma^2(1/2 - \rho)}{L_2}, \frac{6(1 - \rho)}{L_2(s_{\max}^{\text{NC}})^2} \right\}. \quad (8)$$

Theorem 3.3 shows that our gradient scaling is ‘natural,’ as the unit step eventually provides sufficient descent—an attribute typical of Newton-type methods [10, 54]. The bound in (8) is a worst-case result and may differ from typical behavior. In practice, as seen in Section 4, line search usually accepts $\alpha = 1$ for most iterations. Moreover, Appendix B.4 shows that our SPC steps locally satisfy a *curvature condition*, ensuring that the unit step length produces sufficiently large update directions.

Under the standard Lipschitz gradient assumption (with no need for Assumption 3.1), global convergence of our method follows straightforwardly from the application of line search to the gradient, as the scaling factor, s , is bounded below (a consequence of Lemma B.4). However, a more careful analysis reveals that, through our explicit control of the curvature term via (3), we can also guarantee convergence under Assumption 3.1 and a weaker form of gradient Lipschitz smoothness.

Assumption 3.4 (Hessian-gradient Directional Smoothness). There exists $0 \leq L_1 < \infty$ such that for all $\mathbf{x} \in \mathbb{R}^d$, if $\langle \mathbf{g}(\mathbf{x}), \mathbf{H}(\mathbf{x})\mathbf{g}(\mathbf{x}) \rangle > 0$, then $\|\mathbf{H}(\mathbf{x})\mathbf{g}(\mathbf{x})\| \leq L_1 \|\mathbf{g}(\mathbf{x})\|$.

Remark 3.5. For twice continuously differentiable functions, Assumption 3.4 relaxes the standard L_g -Lipschitz gradient smoothness assumption by requiring regularity only along $\mathbf{g}$. Indeed, $L_1 \leq L_g$.

Additionally, the concept of *moral smoothness*, introduced by Roosta et al. [63, Assumption 2], is strictly weaker than Lipschitz gradient and Hessian assumptions on any sublevel set of the gradient norm, yet it still implies Assumption 3.4; see Roosta et al. [63, Lemma 2].

Proposition 3.6 (Global Convergence). *Consider Assumptions 3.1 and 3.4 and suppose f is lower bounded. For any $0 < \varepsilon_g < 1$, after at most $K \in \mathcal{O}(\varepsilon_g^{-2})$ iterations of Algorithm 2, we have $\|\mathbf{g}_k\| \leq \varepsilon_g$ for some $0 \leq k \leq K$.*

The detailed complexity bound in Proposition 3.6, including all underlying constants, is provided in Appendix B. Unsurprisingly, since the search direction is entirely determined by the gradient, the rate in Proposition 3.6 matches that of gradient descent under the Lipschitz gradient smoothness condition with a line search or fixed step size [12, 52]. Furthermore, since our method coincides with the Newton’s method when $d = 1$, it is not surprising that the convergence rate also matches that of Newton’s method in the nonconvex setting with the stronger Lipschitz gradient and Hessian smoothness conditions [12]. The novelty of Proposition 3.6 lies in achieving this complexity under the alternative smoothness conditions, i.e., Assumptions 3.1 and 3.4.

3.2 Local Convergence

Next we consider the local convergence properties of our method. Improved local convergence rates are often considered to be the primary appeal of second order method [54, 64]. Let $\mathbf{x}^*$ be a local minima satisfying the second-order sufficient condition $\mathbf{g}(\mathbf{x}^*) = 0$ and $\mathbf{H}(\mathbf{x}^*) \succ 0$. By the continuity of the Hessian, there exists a ball of radius r around $\mathbf{x}^*$, denoted by $\mathcal{B}_r^*$, such that

$$0 < \mu \triangleq \min_{\mathbf{x} \in \mathcal{B}_r^*} \lambda_{\min}(\mathbf{H}) \leq \max_{\mathbf{x} \in \mathcal{B}_r^*} \lambda_{\max}(\mathbf{H}) \triangleq M < \infty. \quad (9)$$

Our next result demonstrates that, near $\mathbf{x}^*$, the unit step size is acceptable to line search for all iterations, leading to a linear decrease in the sub-optimality of the objective value.

Theorem 3.7. *Consider Assumption 3.1 and suppose $\mathbf{x}^*$ is a local minimum satisfying the second order sufficient conditions. If $\mathbf{x}_0$ is sufficiently close to $\mathbf{x}^*$, then for all iteration of Algorithm 2, the unit step size $\alpha_k = 1$ satisfies the Armijo condition (7) and we have*

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - \tau)(f(\mathbf{x}_k) - f(\mathbf{x}^*)),$$

where $\tau \triangleq 2\rho\mu \max\{1/M, s_{\min}^{\text{LPC}}\} \in (0, 1]$ if $\sigma \geq \mu$, and $\tau \triangleq 2\rho\mu/M \in (0, 1]$ otherwise.

Remark 3.8. Suppose $\mathbf{x} \in \mathcal{B}_r^*$ where $\mathcal{B}_r^*$ is as in (9). Since $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq M \|\mathbf{g}\|^2$, the SPC case only arises on $\mathcal{B}_r^*$ if σ is chosen such that $\sigma \leq M$; otherwise Algorithm 1 always returns the LPC scaling. In this case, since $s_{\min}^{\text{LPC}} < 1/\sigma < 1/M$, and $\rho < 1/2$ we always have $0 < \tau \leq 1$ in Theorem 3.7.

Theorem 3.7 suggests that the rate of convergence for scaled gradient descent has a similar dependence on the local condition number $\kappa \triangleq M/\mu$ as that of the typical gradient descent. However, as the proof of Theorem 3.7 reveals, this worst-case analysis overlooks the potential for scaled gradient methods to exploit larger scalings by adapting to local geometry. For instance, if $\sigma < \mu$ then $s^{\text{SPC}} \leq 1/\mu$. Therefore, in “best-case” iterations where large scalings (close to $1/\mu$) pass the line search with unit step size, the linear rate can be as small as $1 - 2\rho$ (cf. (22)). This rate eliminates dependence on the local condition number, resembling the problem-independent convergence rates of many Newton-type methods [63, 64]. This reflected in the practical performance of our algorithm. In particular, in Section 4, we demonstrate numerically that scaled gradient methods often produce large scalings that pass the line search with unit step size, leading to rapid convergence. Conversely, if $\sigma \gg \mu$, both LPC and SPC scalings are upper bounded by $1/\sigma \ll 1/\mu$, limiting the local rate.

Our next result states that the MR scaling (5) can give rise to linear convergence in the gradient norm. Intuitively, this result arises because the MR scaling minimizes the norm of the residual of the Newton system associated with the scaled gradient, which can also be viewed as the norm of the linearized gradient.

Theorem 3.9. *Consider Assumption 3.1 and suppose $\mathbf{x}^*$ is a local minimum satisfying the second order sufficient conditions. If $\mathbf{x}_0$ is sufficiently close to $\mathbf{x}^*$, then the iterations of the form $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k^{\text{MR}} \mathbf{g}_k$ converge linearly in the gradient norm.*

Our proof of this result (Appendix B.6), extends beyond second-order sufficient conditions to a more general setting. An immediate implication of Theorem 3.9 is that the gradient norm serves

as a “secondary objective” for the MR scaling. This is notable given recent work on gradient norm regularization, which biases gradient descent toward regions with small gradient norms, often linked to “flat” regions and better generalization in machine learning [5, 32, 36, 37, 67, 73]. Unlike explicit regularization, which requires additional hyperparameter tuning, MR scaling implicitly achieves this bias without additional considerations.

4 Numerical Results

We now proceed to validate our results by comparing our scaled gradient method against multiple variants of GD including fixed step size, Nesterov acceleration [52], heavy ball momentum [69], and Adam [39]. In addition, we compare against hyperparameter free methods such as line search GD, with a number of step size resetting techniques inspired by [68], as well as a deterministic variant of Polyak non-monotone (PoNo) Line Search [25] (see Appendix D.2 for details). For visual clarity we report the best performing resetting scheme as “line search” and defer PoNo line search comparison to Appendix D.

For our scaling method, we consider the CG, MR and GM scalings. However, in the quadratic case, it is well known that steepest descent and minimal gradient (our CG and MR resp.) step sizes are prone to a “zig-zag effect” [18, 54]. Dai and Yuan [18] theoretically and empirically demonstrate that this zig-zag effect can be mitigated by alternating between steepest descent and minimal gradient steps. For this reason, we also consider alternating scalings on each SPC iteration⁴. Alternating scalings are denoted by a concatenated string of the two scalings, in order, e.g. “MRCG”.

In the main body, we report the best performing scaling, relegating performance comparison between scalings to Appendix D. Where available, we apply theoretically convergent hyperparameter values for our competitor methods, otherwise we manually tune hyperparameter settings. We note that tuning incurs a significant—and often overlooked—cost. For example, tuning Adam on our logistic regression CIFAR10 task took roughly 20 times longer than a CGMR run.

To fairly compare methods with different per-iteration costs, we plot objective values against the number of *oracle calls*—equivalent function evaluations (see Appendix D.1 for details). For completeness, wall-clock time results are provided in Appendix D. The setup for each problem and optimizer is detailed there as well. All methods are implemented in PyTorch [58] and run on various cluster GPUs (e.g., H100, A100). A link to our code is available in Appendix D.

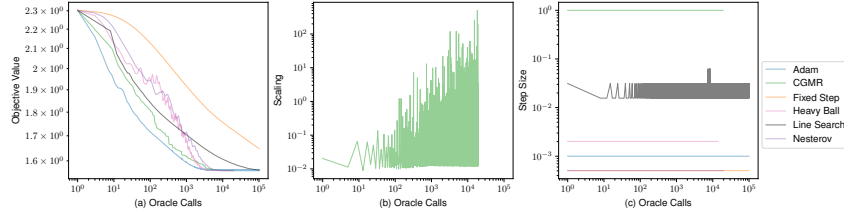


Figure 2: Multi-class logistic regression on CIFAR10. (a) Objective value. (b) Scaling utilized by the CGMR method. (c) Step size.

Multi-class Logistic Regression. In Figure 2, we present multi-class logistic regression with ℓ_2 regularization on the CIFAR10 dataset [40]. This μ -strongly convex problem provides a well-behaved, yet challenging, baseline for comparison. We observe that “CGMR” is competitive with Adam and accelerated methods, while significantly outperforming line search (including PoNo) and fixed step size approaches. This is achieved without requiring any tuning or prior knowledge of problem constants. Notably, the scalings produced by our method oscillate between large (close to $1/\mu$) and small values throughout the optimization trajectory⁵. Despite the magnitude of these scaling values, the line search accepts the unit step length at *all iterations*. In light of Theorem 3.7, we believe these large scaling values may contribute to the rapid convergence of our method.

⁴If an LPC/NC arises, we proceed based on the last SPC step.

⁵As shown in Appendix D.3, this “large scaling” effect is most pronounced for alternating scaling.

Multilayer Perceptron (MLP). In Figure 3, we examine a nonconvex, small MLP on the FashionMNIST dataset [71]. The results show that “CGMR” scaled gradient is again competitive with Adam and significantly outperforms all other methods, all without requiring hyperparameter tuning. Notably, the unit step size is accepted by the line search at almost every SPC iteration. When NC is detected it is exploited using forward tracking search. Meanwhile, LPC does not occur at any point during the optimization trajectory. The scaling chosen by CGMR oscillates between large and small values, mirroring the behavior observed in the convex logistic regression experiment.

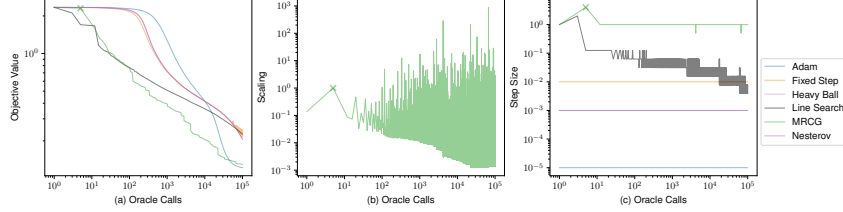


Figure 3: MLP on the FashionMNIST. (a) Objective value (b) Scaling utilized by the CGMR method (c) Step size. Crosses indicate iterations where negative curvature is detected.

ResNet. In Figure 4, we examine an over-parameterized, nonconvex ResNet18 architecture [30] on the Imagenette dataset [33]. The results show that the unit step size is accepted at each iteration for our scaled gradient method, with no NC or LPC directions detected during the iterations. Our method significantly outperforms the only other hyperparameter-free methods (line search and PoNo) but is surpassed by hyperparameter-tuned approaches. In contrast to scaled gradient, the tuned methods exhibit notable instability in the early iterations, followed by stabilization and convergence. This “unstable convergence” effect has been observed to be beneficial in many large-scale, nonconvex models [2, 14]. Given this, it appears that for scaled gradient to be competitive on large-scale problems, some degree of non-monotonicity must be introduced into the iterations. Roulet et al. [65] demonstrate that by using CG-style scaling with large step sizes (e.g., $\alpha \geq 2$), unstable convergence can be induced in a principled manner. A similar approach could be applied to our methods to design *principled* step size schedules, such as annealing from $\alpha \geq 2$ (unstable) to $\alpha = 1$ (stable), to leverage the benefits of the unstable regime. We leave this exploration for future work. Finally, we note that the MR scaling uniformly decreases the gradient with the unit step size across our examples, consistent with Theorem 3.9; see Appendix D .

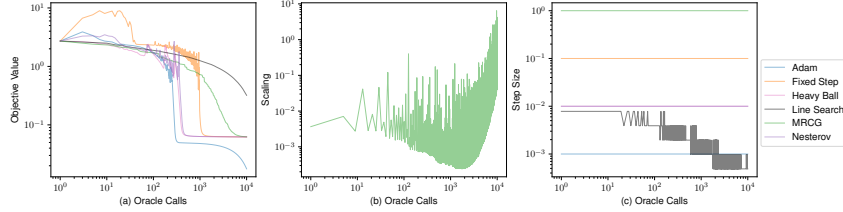


Figure 4: ResNet18 on Imagenette. (a) Objective value. (b) Scaling utilized by the MRGG method. (c) Step size. Crosses indicate iterations where negative curvature is detected.

5 Conclusions

In this work we developed a framework based on (weakened) Lipschitz Hessian smoothness for analyzing Hessian-aware scaling of gradient directions. Under this framework, we show that Hessian curvature information can be used to enhance vanilla GD with a unit step size guarantee. We show this guarantee holds for all iterations near minima satisfying certain regularity conditions. Furthermore, under alternative smoothness conditions, we prove global convergence. Numerically, we observe that the unit step size guarantee holds across most of the optimization trajectory. We also demonstrate that our method outperforms other hyperparameter free algorithms.

Limitations and Future Directions. A limitation of our theory is the local nature of our unit step size guarantee. In particular, Theorem 3.3 suggests that the guarantee may only hold in a region with a

prohibitive dependence on certain problem constants. Fortunately, our numerical results demonstrate that the unit step size guarantee is widely applicable in practice. A natural future direction of work is obtaining theory that more closely tracks the practical performance of our algorithm. This can perhaps be achieved by considering analysis under an alternative globalization framework or with an alternative potential function. Another limitation is performance compared with hand tuned alternatives. Extensions like Hessian aware step size schedules, scaling for adaptive gradient methods, and stochastic variations are promising candidates for addressing this limitation.

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A Additional Material for Section 2

A.1 Discussion of Algorithm 1 Hyperparameters

We now consider how to set the hyperparameters of Algorithm 1, beginning with σ . Recall that in Proposition 2.2, the constant $1/\sigma$ bounds the step size in both the SPC and LPC cases. Furthermore, in Theorem 3.7 we show that better local rates can be attained when σ is smaller than the local strong convexity constant. Together, these results suggest that σ should generally be chosen small ($\sigma \ll 1$). This has the additional effect of maximizing the number of SPC steps, which, due to the second order scaling, tend to be the most effective in practice. Despite the dependence on σ suggested by our local unit step size guarantee (Theorem 3.7), our numerical experience suggests that our algorithm widely utilizes the unit step when σ is small or even zero.

Taking σ to be set small, LPC steps are encountered rarely in practice. Therefore $s_{\min}^{\text{LPC}} = s_{\max}^{\text{LPC}} = s^{\text{LPC}} = 1/\sigma$ is a reasonable choice, even if backtracking is necessary from this large initial step size. Practical experience also suggests that NC directions are relatively uncommon (see Section 4). Moreover, significant progress can be expected when exploiting negative curvature directions with large step sizes [15, 28, 43]. For these reasons, instead of tuning the NC scaling, we suggest setting $s_{\min}^{\text{NC}} = s_{\max}^{\text{NC}} = 1$ and exploiting the NC directions with a forward/back-tracking line search. In the case where the problem is μ -strongly convex, we can take $\sigma = 0$ and disregard s^{LPC} and s^{NC} , since curvature along the gradient direction will always be bounded below by μ . This also implies that, in the strongly convex case, μ can take the place of σ in our analysis.

A.2 Proofs

Proof of Proposition 2.2

Proof. In the LPC and NC cases, the results follow from the definition of s^{LPC} and s^{NC} in Algorithm 1. The SPC condition excludes the case where $\mathbf{g} = 0$, furthermore, it is clear that $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > 0$ implies $\mathbf{H}\mathbf{g} \neq 0$, and hence all of the scalings are well defined. The Cauchy-Schwarz inequality implies $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq \|\mathbf{g}\| \|\mathbf{H}\mathbf{g}\|$ and $\|\mathbf{H}\mathbf{g}\| \geq \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle / \|\mathbf{g}\|$. Applying these two inequalities subsequently to s^{MR} we have

$$0 < s^{\text{MR}} = \frac{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle}{\|\mathbf{H}\mathbf{g}\|^2} \leq \frac{\|\mathbf{g}\|}{\|\mathbf{H}\mathbf{g}\|} \leq \frac{\|\mathbf{g}\|^2}{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle},$$

which captures the three left inequalities in this case. The right-most inequality follows from $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > \sigma \|\mathbf{g}\|^2$ and the definition of (4). $\square$

Proof of Proposition 2.3

Proof. The first-order descent trivially holds. In the NC case, the result immediately follows from $\langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle < 0$ and the first-order descent property. In the LPC case,

$$\begin{aligned} \langle \mathbf{g}, \mathbf{p} \rangle + \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle &= -s^{\text{LPC}} \|\mathbf{g}\|^2 + (s^{\text{LPC}})^2 \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\ &\leq -s^{\text{LPC}} \|\mathbf{g}\|^2 + (s^{\text{LPC}})^2 \sigma \|\mathbf{g}\|^2 \\ &\leq -s^{\text{LPC}} \|\mathbf{g}\|^2 + s^{\text{LPC}} \|\mathbf{g}\|^2 = 0, \end{aligned}$$

where the second to last line follows from the LPC condition and the last line follows from $s^{\text{LPC}} \leq 1/\sigma$ in Proposition 2.2.

For the SPC case, we consider each scaling successively. For the CG scaling, we have

$$\begin{aligned} \langle \mathbf{p}^{\text{CG}}, \mathbf{g} \rangle + \langle \mathbf{p}^{\text{CG}}, \mathbf{H}\mathbf{p}^{\text{CG}} \rangle &= -s^{\text{CG}} \|\mathbf{g}\|^2 + (s^{\text{CG}})^2 \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\ &= -\frac{\|\mathbf{g}\|^4}{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle} + \frac{\|\mathbf{g}\|^4}{(\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle)^2} \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\ &= 0. \end{aligned}$$

For the MR scaling, we have

$$\begin{aligned}
\langle \mathbf{p}^{\text{MR}}, \mathbf{g} \rangle + \langle \mathbf{p}^{\text{MR}}, \mathbf{H}\mathbf{p}^{\text{MR}} \rangle &= -s^{\text{MR}} \|\mathbf{g}\|^2 + (s^{\text{MR}})^2 \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\
&= -\frac{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle}{\|\mathbf{H}\mathbf{g}\|^2} \|\mathbf{g}\|^2 + \frac{(\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle)^2}{\|\mathbf{H}\mathbf{g}\|^4} \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\
&\leq -\frac{\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle}{\|\mathbf{H}\mathbf{g}\|^2} \|\mathbf{g}\|^2 + \frac{\|\mathbf{g}\|^2}{\|\mathbf{H}\mathbf{g}\|^2} \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\
&= 0,
\end{aligned}$$

where the third line follows from the Cauchy-Schwarz inequality. Finally, for the geometric mean scaling, we have

$$\begin{aligned}
\langle \mathbf{p}^{\text{GM}}, \mathbf{g} \rangle + \langle \mathbf{p}^{\text{GM}}, \mathbf{H}\mathbf{p}^{\text{GM}} \rangle &= -s^{\text{GM}} \|\mathbf{g}\|^2 + (s^{\text{GM}})^2 \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\
&= -\frac{\|\mathbf{g}\|^3}{\|\mathbf{H}\mathbf{g}\|} + \frac{\|\mathbf{g}\|^2}{\|\mathbf{H}\mathbf{g}\|^2} \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \\
&\leq -\frac{\|\mathbf{g}\|^3}{\|\mathbf{H}\mathbf{g}\|} + \frac{\|\mathbf{g}\|^2}{\|\mathbf{H}\mathbf{g}\|^2} \|\mathbf{g}\| \|\mathbf{H}\mathbf{g}\| \\
&= -\frac{\|\mathbf{g}\|^3}{\|\mathbf{H}\mathbf{g}\|} + \frac{\|\mathbf{g}\|^3}{\|\mathbf{H}\mathbf{g}\|} = 0,
\end{aligned}$$

where again the third line follows from the Cauchy-Schwarz inequality. □

Proof of Proposition 2.5

Proof. Consider a scalar reparameterization of the original variables $\mathbf{x}$ as $\mathbf{y} = \mathbf{x}/c$, for some $c \neq 0$. Our objective over the new coordinates is $\bar{f}(\mathbf{y}) = f(c\mathbf{y})$ so that

$$\nabla_{\mathbf{y}} \bar{f}(\mathbf{y}) = c \nabla f(\mathbf{x}), \quad \nabla_{\mathbf{y}}^2 \bar{f}(\mathbf{y}) = c^2 \nabla^2 f(\mathbf{x}).$$

Therefore, SPC scalings computed in the transformed coordinates satisfy

$$s(\mathbf{y}) = \frac{s(\mathbf{x})}{c^2}.$$

For example, for the MR scaling we have

$$s^{\text{MR}}(\mathbf{y}) = \frac{\langle \nabla_{\mathbf{y}} \bar{f}(\mathbf{y}), \nabla_{\mathbf{y}}^2 \bar{f}(\mathbf{y}) \nabla_{\mathbf{y}} \bar{f}(\mathbf{y}) \rangle}{\|\nabla_{\mathbf{y}}^2 \bar{f}(\mathbf{y}) \nabla_{\mathbf{y}} \bar{f}(\mathbf{y})\|^2} = \frac{\langle c \nabla f(\mathbf{x}), (c^2 \nabla^2 f(\mathbf{x})) c \nabla f(\mathbf{x}) \rangle}{\|c^2 \nabla^2 f(\mathbf{x}) c \nabla f(\mathbf{x})\|^2} = \frac{1}{c^2} s^{\text{MR}}(\mathbf{x}).$$

Now this implies that, for the SPC scaled gradient directions, the update computed with respect to the new coordinates preserves the reparameterization. Indeed, given $\mathbf{y}_k = \mathbf{x}_k/c$, the scaled gradient descent update at $\mathbf{y}_k$ with some fixed step size $\alpha > 0$ is

$$\mathbf{y}_{k+1} = \mathbf{y}_k - \alpha s(\mathbf{y}_k) \nabla_{\mathbf{y}} \bar{f}(\mathbf{y}_k) = \frac{1}{c} (\mathbf{x}_k - \alpha s(\mathbf{x}_k) \nabla f(\mathbf{x}_k)) = \frac{\mathbf{x}_{k+1}}{c},$$

where $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha s(\mathbf{x}_k) \nabla f(\mathbf{x}_k)$ is the update in the original coordinates. That is, the same relationship between $\mathbf{x}$ and $\mathbf{y}$ holds at the following iteration. □

B Additional Material for Section 3

B.1 Line Search Algorithms

Algorithm 3 Backward Tracking Line Search.

```

1: input: Initial step size  $\alpha^0$ , Scaling parameter  $0 < \theta < 1$ .
2:  $\alpha \leftarrow \alpha^0$ .
3: while (7) is not satisfied do
4:    $\alpha \leftarrow \theta\alpha$ .
5: end while
6: return  $\alpha$ .

```

Algorithm 4 Forward/Backward Tracking Line Search

```

1: input: Initial step size  $\alpha^0$ , Scaling parameter  $0 < \theta < 1$ .
2:  $\alpha \leftarrow \alpha^0$ .
3: if (7) is not satisfied then
4:   Call Algorithm 3
5: else
6:   while (7) is satisfied do
7:      $\alpha = \alpha/\theta$ .
8:   end while
9:   return  $\theta\alpha$ .
10: end if

```

B.2 Derivation and Proof of Theorem 3.3

Throughout the following we routinely make use of the facts, due to collinearity of $\mathbf{g}$ and $\mathbf{p}$, that $\langle \mathbf{p}, \mathbf{g} \rangle = -\|\mathbf{p}\| \|\mathbf{g}\|$, $\|\mathbf{p}\| = s \|\mathbf{g}\|$, and the fact that $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle$ and $\langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle$ have the same sign. Letting $\mathbf{p} = -s\mathbf{g}$, $s \geq 0$, Assumption 3.1 and twice continuous differentiability of f imply

$$\|\mathbf{g}(\mathbf{x} + \mathbf{p}) - \mathbf{g}(\mathbf{x}) - \mathbf{H}(\mathbf{x})\mathbf{p}\| \leq \frac{L_2}{2} \|\mathbf{p}\|^2, \quad (10)$$

$$\left| f(\mathbf{x} + \mathbf{p}) - f(\mathbf{x}) - \langle \mathbf{g}, \mathbf{p} \rangle - \frac{1}{2} \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle \right| \leq \frac{L_2}{6} \|\mathbf{p}\|^3. \quad (11)$$

These bounds can be derived similarly to the general Lipschitz Hessian upper bounds in, e.g., Nesterov and Polyak [53, Lemma 1]. Our analysis is based on applying the second order descent condition (3) to the cubic upper bound in (11). We consider the non-negative and negative curvature cases separately.

Non-negative Curvature Case. Suppose the curvature along the gradient direction is non-negative, i.e., $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \geq 0$. In this case, Algorithm 1 selects either the SPC or LPC scalings. By Propositions 2.2 and 2.3, we have $s \leq 1/\sigma$ and (3). Considering (11) and applying $\alpha \leq 1$ and (3), we have

$$\begin{aligned}
f(\mathbf{x} + \alpha\mathbf{p}) &\leq f(\mathbf{x}) + \alpha \langle \mathbf{g}, \mathbf{p} \rangle + \frac{\alpha^2}{2} \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3 \\
&\leq f(\mathbf{x}) + \frac{\alpha}{2} \langle \mathbf{g}, \mathbf{p} \rangle + \frac{\alpha}{2} (\langle \mathbf{g}, \mathbf{p} \rangle + \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle) + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3 \\
&\leq f(\mathbf{x}) + \frac{\alpha}{2} \langle \mathbf{g}, \mathbf{p} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3.
\end{aligned}$$

Subtracting $f(\mathbf{x}) + \alpha\rho \langle \mathbf{p}, \mathbf{g} \rangle$ from both sides yields

$$\begin{aligned}
f(\mathbf{x} + \alpha\mathbf{p}) - f(\mathbf{x}) - \alpha\rho \langle \mathbf{p}, \mathbf{g} \rangle &\leq \alpha \left(\frac{1}{2} - \rho \right) \langle \mathbf{p}, \mathbf{g} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3 \\
&= -\alpha \left(\frac{1}{2} - \rho \right) \|\mathbf{g}\| \|\mathbf{p}\| + \frac{L_2\alpha^3}{6} s \|\mathbf{g}\| \|\mathbf{p}\|^2 \\
&= \left(\rho - \frac{1}{2} + \frac{L_2\alpha^2 s \|\mathbf{p}\|}{6} \right) \alpha \|\mathbf{p}\| \|\mathbf{g}\| \\
&\leq \left(\rho - \frac{1}{2} + \frac{L_2\alpha^2 \|\mathbf{p}\|}{6\sigma} \right) \alpha \|\mathbf{p}\| \|\mathbf{g}\|,
\end{aligned}$$

where the last inequality follows from $s \leq 1/\sigma$. This implies (7) holds if

$$\alpha \leq \min \left\{ 1, \sqrt{\frac{6\sigma(1/2 - \rho)}{L_2 \|\mathbf{p}\|}} \right\}. \quad (12)$$

Negative Curvature Case. We now consider the case where $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle < 0$, in which case Algorithm 1 selects the NC scaling. As a result, dropping the negative term in (11) gives

$$\begin{aligned}
f(\mathbf{x} + \alpha\mathbf{p}) &\leq f(\mathbf{x}) + \alpha \langle \mathbf{p}, \mathbf{g} \rangle + \frac{\alpha^2}{2} \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3 \\
&\leq f(\mathbf{x}) + \alpha \langle \mathbf{p}, \mathbf{g} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3.
\end{aligned}$$

Similarly to the previous case, subtracting $f(\mathbf{x}) + \alpha\rho \langle \mathbf{p}, \mathbf{g} \rangle$ yields

$$\begin{aligned}
f(\mathbf{x} + \alpha\mathbf{p}) - f(\mathbf{x}) - \alpha\rho \langle \mathbf{p}, \mathbf{g} \rangle &\leq \alpha(1 - \rho) \langle \mathbf{p}, \mathbf{g} \rangle + \frac{L_2\alpha^3}{6} \|\mathbf{p}\|^3 \\
&\leq -\alpha(1 - \rho) \|\mathbf{g}\| \|\mathbf{p}\| + \frac{L_2\alpha^3 s}{6} \|\mathbf{g}\| \|\mathbf{p}\|^2 \\
&\leq \alpha \|\mathbf{p}\| \|\mathbf{g}\| \left(-(1 - \rho) + \frac{L_2\alpha^2 (s_{\max}^{\text{NC}})}{6} \|\mathbf{p}\| \right),
\end{aligned}$$

which implies (7) if

$$\alpha \leq \sqrt{\frac{6(1 - \rho)}{L_2 s_{\max}^{\text{NC}} \|\mathbf{p}\|}}, \quad (13)$$

This analysis demonstrates the importance of separating the non-negative and negative cases. In particular, in the case of the former, the condition $\alpha \leq 1$ is necessary, while in the latter, the second-order term is omitted entirely, allowing for unrestricted step sizes. This distinction explains why large step sizes (e.g., from forward tracking line search) are feasible in the negative curvature case.

The bounds in (12) and (13) ensure that the line search condition is satisfied by small enough step sizes. While these bounds are typically utilized for proving global convergence, we instead establish a *local unit step size* result for our Hessian-aware scaled gradient descent. Specifically, these bounds imply that, for $\|\mathbf{p}\|$ sufficiently small, the line search condition (7) is satisfied with $\alpha = 1$. Through Proposition 2.2, this can be linked to a condition on the gradient, which forms the proof of Theorem 3.3.

Proof of Theorem 3.3. Suppose $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \geq 0$. Since $s \leq 1/\sigma$, we have $\|\mathbf{p}\| \leq \|\mathbf{g}\|/\sigma$ so that if (8) holds then

$$\|\mathbf{p}\| \leq \frac{\|\mathbf{g}\|}{\sigma} \leq \frac{6\sigma(1/2 - \rho)}{L_2},$$

which implies (12) holds with $\alpha = 1$. When $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle < 0$, we have $s \leq s_{\max}^{\text{NC}}$, and hence if (8) then

$$\|\mathbf{p}\| \leq s_{\max}^{\text{NC}} \|\mathbf{g}\| \leq \frac{6(1 - \rho)}{L_2 s_{\max}^{\text{NC}}},$$

which implies (13) with $\alpha = 1$. $\square$

B.3 Proof of Proposition 3.6

Let us first restate Proposition 3.6 with all the details and underlying constants.

Proposition B.1 (Restatement of Proposition 3.6). *Consider Assumptions 3.1 and 3.4 and suppose $-\infty < f^* = \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$. For any $0 < \varepsilon_g < 1$, after at most*

$$K = \left\lceil \frac{f(\mathbf{x}_0) - f^*}{\min\{c^{SPC}, c^{LPC}, c^{NC}\} \varepsilon_g^2} \right\rceil,$$

iterations of Algorithm 2, we have $\|\mathbf{g}_k\| \leq \varepsilon_g$, where c^{SPC} , c^{LPC} and c^{NC} are defined as follows:

$$\begin{aligned} c^{SPC} &\triangleq \sigma \rho \min \left\{ \left(\frac{6\sigma(\frac{1}{2} - \rho)}{L_2} \right)^2, \frac{1}{L_1^2} \right\} \\ c^{LPC} &\triangleq \rho \min \left\{ s_{\min}^{LPC}, \sqrt{\frac{6\sigma(\frac{1}{2} - \rho)s_{\min}^{LPC}}{L_2}} \right\}, \\ c^{NC} &\triangleq \rho \sqrt{\frac{6(1 - \rho)s_{\min}^{NC}}{L_2 s_{\max}^{NC}}}. \end{aligned}$$

Before providing the proof, we note that, as discussed earlier, L_1 and L_2 are lower bounds for the global Lipschitz constants of the problem (if such constants exist), so the factor based on L_1 and L_2 could be smaller than those based on their global counterparts. The constant factor in Proposition B.1 depends quadratically on both L_1 and L_2 , but the dependence on L_1 can be improved to linear if the MR scaling is not used; see Remark B.5. This would align with the dependence on the Lipschitz gradient constant for standard gradient descent in the nonconvex setting.

To prove Proposition B.1, we first consider a lemma which analyzes the worst case *per-iteration* descent for the NC, LPC, and SPC separately.

Lemma B.2 (Per-iteration Descent). *Consider Assumption 3.1, and let $\mathbf{p}$ and $\alpha > 0$ be generated by Algorithm 2. If $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle < 0$ (NC), then*

$$f(\mathbf{x} + \alpha \mathbf{p}) \leq f(\mathbf{x}) - \rho \sqrt{\frac{6(1 - \rho)s_{\min}^{NC}}{L_2 s_{\max}^{NC}}} \|\mathbf{g}\|^{3/2}.$$

If $0 \leq \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq \sigma \|\mathbf{g}\|^2$ (LPC), then

$$f(\mathbf{x} + \alpha \mathbf{p}) \leq f(\mathbf{x}) - \rho \min \left\{ s_{\min}^{LPC} \|\mathbf{g}\|^2, \sqrt{\frac{6\sigma(1/2 - \rho)s_{\min}^{LPC}}{L_2}} \|\mathbf{g}\|^{3/2} \right\}.$$

Suppose $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle > \sigma \|\mathbf{g}\|^2$ (SPC). If

$$\|\mathbf{p}\| \geq \frac{6\sigma(1/2 - \rho)}{L_2}, \tag{14}$$

then

$$f(\mathbf{x} + \alpha \mathbf{p}) \leq f(\mathbf{x}) - \rho \sigma \left(\frac{6\sigma(1/2 - \rho)}{L_2} \right)^2.$$

Otherwise, we have $\alpha = 1$ and

$$f(\mathbf{x} + \mathbf{p}) \leq f(\mathbf{x}) - \rho \|\mathbf{p}\| \|\mathbf{g}\|.$$

Proof. (NC) From (13), it follows that the largest step size satisfying (7) must satisfy

$$\alpha \geq \sqrt{\frac{6(1 - \rho)}{L_2 s_{\max}^{NC} \|\mathbf{p}\|}}.$$

Hence,

$$f(\mathbf{x} + \alpha \mathbf{p}) - f(\mathbf{x}) \leq \rho \alpha \langle \mathbf{p}, \mathbf{g} \rangle = -\rho \alpha \|\mathbf{p}\| \|\mathbf{g}\| \leq -\rho \sqrt{\frac{6(1-\rho) \|\mathbf{p}\|}{L_2 s_{\max}^{\text{NC}}}} \|\mathbf{g}\|.$$

Applying $\|\mathbf{p}\| \geq s_{\min}^{\text{NC}} \|\mathbf{g}\|$ gives the result.

(LPC) Similarly, (12) implies that the largest step size $\alpha \in (0, 1]$ satisfying (7) must satisfy

$$\alpha \geq \min \left\{ 1, \sqrt{\frac{6\sigma(1/2 - \rho)}{L_2 \|\mathbf{p}\|}} \right\}.$$

Now, from (7) we obtain

$$\begin{aligned} f(\mathbf{x} + \alpha \mathbf{p}) - f(\mathbf{x}) &\leq \rho \alpha \langle \mathbf{p}, \mathbf{g} \rangle = -\rho \alpha \|\mathbf{p}\| \|\mathbf{g}\| \\ &\leq -\rho \min \left\{ \|\mathbf{p}\| \|\mathbf{g}\|, \sqrt{\frac{6\sigma(1/2 - \rho) \|\mathbf{p}\|}{L_2}} \|\mathbf{g}\| \right\}. \end{aligned}$$

Since $s^{\text{LPC}} \geq s_{\min}^{\text{LPC}}$, $\|\mathbf{p}\| \geq s_{\min}^{\text{LPC}} \|\mathbf{g}\|$, which yields the result.

(SPC) Similar to the LPC case, (12) implies that the largest step size $\alpha \in (0, 1]$ satisfying (7) must satisfy

$$\min \left\{ 1, \sqrt{\frac{6\sigma(1/2 - \rho)}{L_2 \|\mathbf{p}\|}} \right\} \leq \alpha \leq 1.$$

Suppose (14) holds, which implies

$$\alpha \geq \sqrt{\frac{6\sigma(1/2 - \rho)}{L_2 \|\mathbf{p}\|}}.$$

In light of (3) and (7) and the fact that $\langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle > \sigma \|\mathbf{p}\|^2$, we have

$$f(\mathbf{x} + \alpha \mathbf{p}) - f(\mathbf{x}) \leq \rho \alpha \langle \mathbf{p}, \mathbf{g} \rangle \leq -\rho \alpha \langle \mathbf{p}, \mathbf{H}\mathbf{p} \rangle \leq -\rho \alpha \sigma \|\mathbf{p}\|^2 \leq -\rho \sigma \left(\frac{6\sigma(1/2 - \rho)}{L_2} \right)^2,$$

where the final inequality follows from (14). If (14) does not hold, then $\alpha = 1$ and hence

$$f(\mathbf{x} + \mathbf{p}) - f(\mathbf{x}) \leq \rho \langle \mathbf{p}, \mathbf{g} \rangle = -\rho \|\mathbf{g}\| \|\mathbf{p}\|.$$

□

Remark B.3. Many of the cases considered in Lemma B.2 suggest descent with an improved dependence on the gradient norm, when compared with standard gradient descent. For instance, in the NC case the dependence is $\|\mathbf{g}\|^{3/2}$, which matches the optimal rate for second order methods in the nonconvex setting. On the other hand, in the SPC case, with large $\|\mathbf{p}\|$, the descent is *independent* of the gradient norm. An unfortunate consequence of worst case analysis is that these improvements are not reflected in the final rate.

By inspecting the proof of Lemma B.2, we see that the per iteration decrease is independent of the scaling magnitude in all cases except SPC steps with $\alpha = 1$. As discussed in the main body, we handle this case with a regularity condition on the Hessian-gradient product, Assumption 3.4. In particular, by combining (5), $\sigma \|\mathbf{g}\|^2 < \langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle$, and $\|\mathbf{H}\mathbf{g}\| \leq L_1 \|\mathbf{g}\|$ from Assumption 3.4, we can obtain a lower bound on the SPC scalings.

Lemma B.4 (Scaling Lower Bounds). *Consider Assumption 3.4 and let $\sigma \leq L_1$. We have*

$$\frac{\sigma}{L_1^2} \leq s^{\text{MR}} \leq s^{\text{GM}} \leq s^{\text{CG}}. \quad (15)$$

Remark B.5. Note that since $\langle \mathbf{g}, \mathbf{H}\mathbf{g} \rangle \leq \|\mathbf{H}\mathbf{g}\| \|\mathbf{g}\| \leq L_1 \|\mathbf{g}\|^2$, the SPC case is detected only when σ is chosen such that $\sigma \leq L_1$; otherwise Algorithm 1 always returns either of LPC or NC scalings. Additionally, by applying $\|\mathbf{H}\mathbf{g}\| \leq L_1 \|\mathbf{g}\|$ directly to s^{GM} we can sharpen the lower bound for the CG and GM scalings to $1/L_1 \leq s^{\text{GM}} \leq s^{\text{CG}}$.

The proof of Proposition B.1 proceeds by combining the worst case analysis of Lemma B.2 with the lower bounds from Lemma B.4.

Proof of Proposition B.1. Suppose that $\|\mathbf{g}_k\| \leq \varepsilon_g$ fails to hold for $k = 0, \dots, K-1$. We divide the iterations $\mathcal{K} = \{0, \dots, K-1\}$ into a disjoint union $\mathcal{K} = \mathcal{K}_{\text{NC}} \cup \mathcal{K}_{\text{LPC}} \cup \mathcal{K}_{\text{SPC}}$ where

$$\begin{aligned}\mathcal{K}_{\text{NC}} &= \{k \in \mathcal{K} \mid \langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle < 0\} \\ \mathcal{K}_{\text{LPC}} &= \left\{k \in \mathcal{K} \mid 0 \leq \langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle \leq \sigma \|\mathbf{g}_k\|^2\right\} \\ \mathcal{K}_{\text{SPC}} &= \left\{k \in \mathcal{K} \mid \langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle > \sigma \|\mathbf{g}_k\|^2\right\}.\end{aligned}$$

For $k \in \mathcal{K}_{\text{NC}}$, Lemma B.2, $\|\mathbf{g}_k\| > \varepsilon_g$ and $\varepsilon_g < 1$ give

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) > \rho \sqrt{\frac{6(1-\rho)s_{\min}^{\text{NC}}}{L_2 s_{\max}^{\text{NC}}}} \varepsilon_g^{3/2} \geq c^{\text{NC}} \varepsilon_g^2,$$

Similarly, for $k \in \mathcal{K}_{\text{LPC}}$, Lemma B.2, $\|\mathbf{g}_k\| > \varepsilon_g$, and $\varepsilon_g < 1$ imply

$$\begin{aligned}f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) &> \rho \min \left\{ s_{\min}^{\text{LPC}} \|\mathbf{g}_k\|^2, \sqrt{\frac{6\sigma(1/2-\rho)s_{\min}^{\text{LPC}}}{L_2}} \|\mathbf{g}_k\|^{3/2} \right\} \\ &> \rho \min \left\{ s_{\min}^{\text{LPC}} \varepsilon_g^2, \sqrt{\frac{6\sigma(1/2-\rho)s_{\min}^{\text{LPC}}}{L_2}} \varepsilon_g^{3/2} \right\} \\ &\geq c^{\text{LPC}} \varepsilon_g^2,\end{aligned}$$

For $k \in \mathcal{K}_{\text{SPC}}$, if (14) does not hold, then Lemmas B.2 and B.4 give

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq \rho \|\mathbf{p}_k\| \|\mathbf{g}_k\| \geq \frac{\sigma \rho}{L_1^2} \|\mathbf{g}_k\|^2 > \frac{\sigma \rho}{L_1^2} \varepsilon_g^2.$$

We can combine this with the case where (14) is satisfied to obtain

$$\begin{aligned}f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) &> \rho \min \left\{ \sigma \left(\frac{6\sigma^2(1/2-\rho)}{L_2} \right)^2, \frac{\sigma}{L_1^2} \varepsilon_g^2 \right\} \\ &\geq \sigma \rho \min \left\{ \left(\frac{6\sigma(1/2-\rho)}{L_2} \right)^2, \frac{1}{L_1^2} \right\} \varepsilon_g^2 \\ &= c^{\text{SPC}} \varepsilon_g^2,\end{aligned}$$

where in the second line we applied $\varepsilon_g < 1$. Finally, we apply a telescoping sum over each case

$$\begin{aligned}f(\mathbf{x}_0) - f(\mathbf{x}_K) &= \sum_{i=0}^{K-1} f(\mathbf{x}_i) - f(\mathbf{x}_{i+1}) \\ &= \sum_{k \in \mathcal{K}_{\text{NC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) + \sum_{k \in \mathcal{K}_{\text{LPC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) + \sum_{k \in \mathcal{K}_{\text{SPC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \\ &> |\mathcal{K}_{\text{NC}}| c^{\text{NC}} \varepsilon_g^2 + |\mathcal{K}_{\text{LPC}}| c^{\text{LPC}} \varepsilon_g^2 + |\mathcal{K}_{\text{SPC}}| c^{\text{SPC}} \varepsilon_g^2 \\ &\geq (|\mathcal{K}_{\text{NC}}| + |\mathcal{K}_{\text{LPC}}| + |\mathcal{K}_{\text{SPC}}|) \min \{c^{\text{NC}}, c^{\text{LPC}}, c^{\text{SPC}}\} \varepsilon_g^2 \\ &= K \min \{c^{\text{NC}}, c^{\text{LPC}}, c^{\text{SPC}}\} \varepsilon_g^2 \\ &\geq f(\mathbf{x}_0) - f^*,\end{aligned}$$

which implies $f^* > f(\mathbf{x}_K)$, leading to a contradiction. $\square$

B.4 Curvature Condition Extension

Our result in Theorem 3.7 offers a guarantee on when $\alpha = 1$ will satisfy the Amijo condition (7). Recall that Amijo condition will be satisfied by any sufficiently short step length (if $\alpha = 1$ any small scaling s). For this reason the Amijo condition is often paired with a *curvature condition* [54], which ensures that the step length is *large enough*, in the sense that the directional derivative at the new point has increased sufficiently from its initial value. That is, for a descent direction $\mathbf{p}$ and $\eta \in (0, 1)$ we require α to satisfy

$$\langle \mathbf{g}(\mathbf{x} + \alpha \mathbf{p}), \mathbf{p} \rangle \geq \eta \langle \mathbf{g}(\mathbf{x}), \mathbf{p} \rangle. \quad (16)$$

When $\rho < \eta$ the pair (7) and (16) are known as the Wolfe conditions. It should be noted that, while the Amijo condition (7) can be guaranteed to hold by employing backtracking line search, satisfaction of the Wolfe conditions requires more sophisticated search techniques [54]. Fortunately, the next proposition reveals that, under certain smoothness conditions, SPC scalings automatically satisfy (16) locally with $\alpha = 1$.

Proposition B.6. *Consider Assumption 3.1. If*

$$\|\mathbf{g}\| \leq \frac{2\sigma^2\eta}{L_2}, \quad (17)$$

then (16) is satisfied by $\mathbf{p}^{CG}$ with $\alpha = 1$ and $\eta \in (0, 1)$. Now consider, in addition, Assumption 3.4. If

$$\|\mathbf{g}\| \leq \frac{2\sigma^2}{L_2} \left(\eta - \left(1 - \frac{\sigma^2}{L_1^2} \right) \right), \quad (18)$$

then $\mathbf{p}^{GM}$ and $\mathbf{p}^{MR}$ satisfy (16) with $\alpha = 1$ and $\eta > 1 - \sigma^2/L_1^2$.

Proof of Proposition B.6. Note that in this proof we make explicit reference to the iteration counter to make tracking updates simpler. In particular, we write $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{p}_k$. A Taylor expansion for the gradient yields

$$\begin{aligned} \mathbf{g}_{k+1} &= \mathbf{g}_k + \int_0^1 \mathbf{H}(\mathbf{x}_k + t\mathbf{p}_k) \mathbf{p}_k dt \\ &= \mathbf{g}_k + \int_0^1 (\mathbf{H}(\mathbf{x}_k + t\mathbf{p}_k) - \mathbf{H}_k + \mathbf{H}_k) \mathbf{p}_k dt. \end{aligned}$$

Taking the inner product with $\mathbf{p}_k$ and applying Assumption 3.1

$$\begin{aligned} \langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle &= \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle + \int_0^1 \langle \mathbf{p}_k, (\mathbf{H}(\mathbf{x}_k + t\mathbf{p}_k) - \mathbf{H}_k) \mathbf{p}_k \rangle dt \\ &\geq \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle - L_2 \int_0^1 t \|\mathbf{p}_k\|^3 dt \\ &= \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle - \frac{L_2}{2} \|\mathbf{p}_k\|^3. \end{aligned}$$

We can write $\|\mathbf{p}_k\|^3 = -s_k \langle \mathbf{g}_k, \mathbf{p}_k \rangle \|\mathbf{p}_k\|$ so that

$$\langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle \geq \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle + \frac{L_2}{2} s_k \langle \mathbf{g}_k, \mathbf{p}_k \rangle \|\mathbf{p}_k\|.$$

Now considering the CG scaling specifically. Combining the condition on the gradient norm (17) with Proposition 2.2 we have

$$s_k^{CG} \|\mathbf{p}_k^{CG}\| \leq \frac{\|\mathbf{g}_k\|}{\sigma^2} \leq \frac{2\eta}{L_2},$$

so that

$$\langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle \geq \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \langle \mathbf{p}_k, \mathbf{H}_k \mathbf{p}_k \rangle + \eta \langle \mathbf{g}_k, \mathbf{p}_k \rangle.$$

Finally, for the CG scaling, $\langle \mathbf{g}_k, \mathbf{p}_k^{CG} \rangle + \langle \mathbf{p}_k^{CG}, \mathbf{H}_k \mathbf{p}_k^{CG} \rangle = 0$ (see the proof of Proposition 2.3 in Appendix A) so

$$\langle \mathbf{g}_{k+1}, \mathbf{p}_k^{CG} \rangle \geq \eta \langle \mathbf{g}_k, \mathbf{p}_k^{CG} \rangle,$$

which is (16).

For the GM and MR scalings the second order descent condition (3) doesn't hold with equality, so we need to lower bound the second order expansion. Starting from the positive curvature test $\langle \mathbf{p}_k, \mathbf{H}\mathbf{p}_k \rangle \geq \sigma \|\mathbf{p}_k\|^2$ we have

$$\begin{aligned} \langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle &\geq \langle \mathbf{g}_k, \mathbf{p}_k \rangle + \sigma \|\mathbf{p}_k\|^2 + \frac{L_2}{2} s_k \langle \mathbf{g}_k, \mathbf{p}_k \rangle \|\mathbf{p}_k\| . \\ &= \left(1 - s_k \sigma + \frac{L_2}{2} s_k \|\mathbf{p}_k\| \right) \langle \mathbf{g}_k, \mathbf{p}_k \rangle . \end{aligned}$$

Next we apply a lower bound to the scaling, which requires Assumption 3.4. In particular, from Lemma B.4 we have $s^{\text{GM}} \geq s^{\text{MR}} \geq \sigma/L_1^2$ and so for $\mathbf{p}_k = -s_k \mathbf{g}_k$ with $s_k \in \{s^{\text{GM}}, s^{\text{MR}}\}$ we have

$$\langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle \geq \left(1 - \sigma^2/L_1^2 + \frac{L_2}{2} s_k \|\mathbf{p}_k\| \right) \langle \mathbf{g}_k, \mathbf{p}_k \rangle .$$

Finally, applying (18) and Proposition 2.2 we obtain

$$s_k \|\mathbf{p}_k\| \leq \frac{\|\mathbf{g}_k\|}{\sigma^2} \leq \frac{2}{L_2} \left(\eta - \left(1 - \frac{\sigma^2}{L_1^2} \right) \right) ,$$

and hence $\langle \mathbf{g}_{k+1}, \mathbf{p}_k \rangle \geq \eta \langle \mathbf{g}_k, \mathbf{p}_k \rangle$. $\square$

We remark that an improved dependence on σ and L_1 can be obtained in (18) for the GM scaling by applying the tighter bound $s_k^{\text{GM}} \geq 1/L_1$ (See Remark B.5).

B.5 Proof of Theorem 3.7

We begin with a lemma, which bounds on the objective function sub-optimality and the gradient norm in a region local to a second order sufficient minima.

Lemma B.7. *Suppose that $\mathbf{x}^*$ satisfies the second order sufficient conditions. Furthermore, suppose that r is chosen such that (9) holds for any $\mathbf{x} \in \mathcal{B}_r^*$, then*

$$\frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|^2 \leq f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{M}{2} \|\mathbf{x} - \mathbf{x}^*\|^2 , \quad (19)$$

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{1}{2\mu} \|\mathbf{g}\|^2 , \quad (20)$$

$$\mu \|\mathbf{x} - \mathbf{x}^*\| \leq \|\mathbf{g}(\mathbf{x})\| \leq M \|\mathbf{x} - \mathbf{x}^*\| . \quad (21)$$

Proof. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, the mean value theorem for twice differentiable functions implies

$$f(\mathbf{y}) = f(\mathbf{x}) + \langle \mathbf{g}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{1}{2} \langle \mathbf{y} - \mathbf{x}, \mathbf{H}(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) \rangle ,$$

for some $t \in (0, 1)$. If $\mathbf{x}, \mathbf{y} \in \mathcal{B}_r^*$, then $\mathbf{x} + t(\mathbf{y} - \mathbf{x}) \in \mathcal{B}_r^*$, and so

$$\frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 \leq f(\mathbf{y}) - f(\mathbf{x}) - \langle \mathbf{g}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \leq \frac{M}{2} \|\mathbf{y} - \mathbf{x}\|^2 .$$

By setting $\mathbf{x} = \mathbf{x}^*$ and $\mathbf{y} = \mathbf{x} \in \mathcal{B}_r^*$ we obtain (19). Now, by rearranging we obtain

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{g}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 .$$

Minimizing both sides over $\mathcal{B}_r^*$ gives (20) since

$$\begin{aligned} f(\mathbf{x}^*) &= \min_{\mathbf{y} \in \mathcal{B}_r^*} f(\mathbf{y}) \geq \min_{\mathbf{y} \in \mathcal{B}_r^*} f(\mathbf{x}) + \langle \mathbf{g}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 \\ &\geq \min_{\mathbf{y} \in \mathbb{R}^d} f(\mathbf{x}) + \langle \mathbf{g}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 \\ &= f(\mathbf{x}) - \frac{1}{2\mu} \|\mathbf{g}(\mathbf{x})\|^2 . \end{aligned}$$

Finally, recall that for twice continuously differentiable functions we have

$$\mathbf{g}(\mathbf{x}) = \int_0^1 \mathbf{H}(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*) dt.$$

Hence, for $\mathbf{x} \in \mathcal{B}_r^*$, we get

$$\begin{aligned} \|\mathbf{g}(\mathbf{x})\|^2 &= \langle \mathbf{g}(\mathbf{x}), \mathbf{g}(\mathbf{x}) \rangle = \left\langle \int_0^1 \mathbf{H}(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*) dt, \int_0^1 \mathbf{H}(\mathbf{x}^* + s(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*) ds \right\rangle \\ &= \int_0^1 \int_0^1 \langle \mathbf{H}(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*), \mathbf{H}(\mathbf{x}^* + s(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*) \rangle ds dt \\ &= \int_0^1 \int_0^1 \langle \mathbf{x} - \mathbf{x}^*, \mathbf{H}(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)) \mathbf{H}(\mathbf{x}^* + s(\mathbf{x} - \mathbf{x}^*))(\mathbf{x} - \mathbf{x}^*) \rangle ds dt. \end{aligned}$$

For positive definite matrices $\mathbf{A}$ and $\mathbf{B}$, we have $\lambda_{\min}(\mathbf{A})\lambda_{\min}(\mathbf{B}) \leq \lambda_{\min}(\mathbf{AB})$ and $\lambda_{\max}(\mathbf{AB}) \leq \lambda_{\max}(\mathbf{A})\lambda_{\max}(\mathbf{B})$. Now since $t, s \in [0, 1]$ we have

$$\mu^2 \mathbf{I} \preceq \mathbf{H}(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)) \mathbf{H}(\mathbf{x}^* + s(\mathbf{x} - \mathbf{x}^*)) \preceq M^2 \mathbf{I},$$

which gives the bounds in (21). $\square$

With Lemma B.7 in hand we prove Theorem 3.7.

Proof of Theorem 3.7. Since $\mathbf{g}(\mathbf{x}^*) = 0$ and the gradient is continuous, we can choose $r' \leq r$ small enough that (8) and (9) hold on $\mathbf{x} \in \mathcal{B}_{r'}^*$. Now suppose that $\mathbf{x}_k \in \mathcal{B}_{r'}^*$, Theorem 3.3 and (7) and (20) imply

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) + \rho \langle \mathbf{p}_k, \mathbf{g}_k \rangle = f(\mathbf{x}_k) - s_k \rho \|\mathbf{g}_k\|^2 \leq f(\mathbf{x}_k) - 2s_k \mu \rho (f(\mathbf{x}_k) - f(\mathbf{x}^*)),$$

which implies

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - 2\mu \rho s_k) (f(\mathbf{x}_k) - f(\mathbf{x}^*)). \quad (22)$$

Since $\mathbf{H}_k \succ 0$, we only need to consider LPC and SPC scalings. If $\mathbf{g}_k$ is a SPC direction, then following a similar line of reasoning as that in Lemma B.4, with M taking the place of L_1 , we get $s_k = s_{\text{SPC}}^{\text{SPC}} \geq 1/M$. Note that for the MR scaling, this bound follows from $\mu \mathbf{I} \preceq \mathbf{H}_k \preceq M \mathbf{I}$ and Lemma C.1. Hence, (22) gives

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq \left(1 - \frac{2\mu \rho}{M}\right) (f(\mathbf{x}_k) - f(\mathbf{x}^*)). \quad (23)$$

If $\sigma < \mu$ then clearly $\mathbf{g}_k$ cannot be an LPC direction and so (23) yields the desired recursion in function sub-optimality. On the other hand, if $\sigma \geq \mu$ and $\mathbf{g}_k$ is an LPC direction, then we simply apply $s_{\min}^{\text{LPC}} \leq s^{\text{LPC}}$ to (22) to obtain

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - 2\mu s_{\min}^{\text{LPC}} \rho) (f(\mathbf{x}_k) - f(\mathbf{x}^*)). \quad (24)$$

The desired recursion follows from combining (23) and (24).

It remains to show that the iterates remain in $\mathcal{B}_{r'}^*$. To this end, define the set

$$\mathcal{F} = \left\{ \mathbf{x} \mid f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{\mu \sigma^2 r^2}{2(\sigma + M)^2} \right\}.$$

Now suppose that an iterate $\mathbf{x}_k$ is sufficiently close to $\mathbf{x}^*$ such that $\mathbf{x}_k \in \mathcal{B}_{r'}^* \cap \mathcal{F}$. Such $\mathbf{x}_k$ exists, as implied by the upper bound in (19). We now show that all subsequent iterates remain in $\mathcal{B}_{r'}^* \cap \mathcal{F}$. Indeed, from the earlier part of the proof, we get $f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - \tau)(f(\mathbf{x}_k) - f(\mathbf{x}^*))$, which implies $\mathbf{x}_{k+1} \in \mathcal{F}$. On the other hand, from $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k \mathbf{g}_k$ (since $\alpha = 1$ is accepted by the line search), $s \leq 1/\sigma$ (Proposition 2.2), the upper bound in (21), and the lower bound in (19), it follows that

$$\begin{aligned} \|\mathbf{x}_{k+1} - \mathbf{x}^*\| &\leq \|\mathbf{x}_k - \mathbf{x}^*\| + s_k \|\mathbf{g}_k\| \leq \left(1 + \frac{M}{\sigma}\right) \|\mathbf{x}_k - \mathbf{x}^*\| \\ &\leq \left(1 + \frac{M}{\sigma}\right) \sqrt{\frac{2}{\mu} (f(\mathbf{x}_k) - f(\mathbf{x}^*))} \leq r, \end{aligned}$$

which gives $\mathbf{x}_{k+1} \in \mathcal{B}_{r'}^*$, and hence $\mathbf{x}_{k+1} \in \mathcal{B}_{r'}^* \cap \mathcal{F}$. $\square$

B.6 Convergence in Gradient Norm and Proof of Theorem 3.9

We prove Theorem 3.9 by considering a more general result, Proposition B.11, to which Theorem 3.9 is a corollary. Our analysis begins by utilizing Assumption 3.1 and a bound on the residual to obtain a recursion in the gradient norm.

Lemma B.8. *Considering Assumption 3.1 and supposing $\|\mathbf{Hg}\| \neq 0$ and $\|\mathbf{g}\| \neq 0$ we have the following recursion for the gradient with the step $\mathbf{p} = -s^{\text{MR}}\mathbf{g}$*

$$\|\mathbf{g}(\mathbf{x} + \mathbf{p})\| \leq \frac{L_2}{2} \|\mathbf{p}\|^2 + \sqrt{1 - \cos^2 \theta} \|\mathbf{g}\|, \quad (25)$$

where $\theta \triangleq \angle(\mathbf{g}, \mathbf{Hg})$ is the angle between $\mathbf{g}$ and $\mathbf{Hg}$.

Proof. Adding and subtracting appropriate values and applying the triangle inequality we have

$$\begin{aligned} \|\mathbf{g}(\mathbf{x} + \mathbf{p})\| &= \|\mathbf{g}(\mathbf{x} + \mathbf{p}) + \mathbf{g} + \mathbf{Hp} - \mathbf{g} - \mathbf{Hp}\| \\ &\leq \|\mathbf{g}(\mathbf{x} + \mathbf{p}) - \mathbf{g} - \mathbf{Hp}\| + \|\mathbf{g} + \mathbf{Hp}\| \\ &= \|\mathbf{g}(\mathbf{x} + \mathbf{p}) - \mathbf{g} - \mathbf{Hp}\| + \|\mathbf{r}\|. \end{aligned}$$

Now, Assumption 3.1 and (10) gives

$$\|\mathbf{g}(\mathbf{x} + \mathbf{p})\| \leq \frac{L_2}{2} \|\mathbf{p}\|^2 + \|\mathbf{r}\|.$$

We can handle the residual term using the properties of MR scaling. Indeed, applying the definition of s^{MR} for $\mathbf{Hg} \neq 0$ we have

$$\begin{aligned} \|\mathbf{r}\|^2 &= \|\mathbf{g} - s^{\text{MR}}\mathbf{Hg}\|^2 = \|\mathbf{g}\|^2 - 2s^{\text{MR}} \langle \mathbf{g}, \mathbf{Hg} \rangle + (s^{\text{MR}})^2 \|\mathbf{Hg}\|^2 \\ &= \|\mathbf{g}\|^2 - 2 \frac{\langle \mathbf{g}, \mathbf{Hg} \rangle^2}{\|\mathbf{Hg}\|^2} + \frac{\langle \mathbf{g}, \mathbf{Hg} \rangle^2}{\|\mathbf{Hg}\|^2} \\ &= \|\mathbf{g}\|^2 \left(1 - \frac{\langle \mathbf{g}, \mathbf{Hg} \rangle^2}{\|\mathbf{Hg}\|^2 \|\mathbf{g}\|^2} \right) \\ &= \|\mathbf{g}\|^2 (1 - \cos^2 \theta), \end{aligned}$$

Putting this all together yields

$$\|\mathbf{g}(\mathbf{x} + \mathbf{p})\| \leq \frac{L_2}{2} \|\mathbf{p}\|^2 + \sqrt{1 - \cos^2 \theta} \|\mathbf{g}\|.$$

which is (25). $\square$

To apply (25) for convergence in gradient norm, the magnitude of $(s^{\text{MR}})^2$ needs to be bounded from above and $\cos \theta$ must remain bounded away from zero. To that end, we introduce Assumption B.9, which excludes cases where the MR scaling becomes unbounded and $\theta \approx \pi/2$, meaning $\mathbf{Hg}$ and $\mathbf{g}$ become nearly orthogonal.

Assumption B.9. Given $\mathbf{x}_0$, there exists constants $\nu_0 > 0$ and $\mu_0 > 0$, possibly depending on $\mathbf{x}_0$, such that for all the iterates of the form $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k^{\text{MR}}\mathbf{g}_k$, $k \geq 0$, with $\mathbf{g}_k \neq 0$ we have

$$\mu_0 \|\mathbf{g}_k\|^2 \leq |\langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle|, \quad (26)$$

$$\nu_0 \leq \cos^2 \theta_k. \quad (27)$$

Remark B.10. The conditions in Assumption B.9 are general and somewhat unconventional. However, as we shall see they are a weakening of more typical conditions. Indeed, any smooth and strongly convex function on the set containing the iterates satisfies Assumption B.9. Furthermore, as we shall see in the proof of Theorem 3.9, the second order sufficient conditions subsume Assumption B.9 for $\mathbf{x}_0$ close enough to a minima. The gradient curvature condition, (26), is a weakening of strong convexity, as the curvature along gradient direction need only be uniformly bounded away from zero (instead of uniformly positive). Applying the Cauchy-Schwarz inequality to (26) we see that $\|\mathbf{g}_k\| > 0$ implies $\|\mathbf{H}_k \mathbf{g}_k\| > 0$, which ensures that $\cos \theta_k$ and s_k^{MR} remain well defined along the path of the iterates. On the other hand, the cosine condition (27) is implied by (26) along with a bound of the form $\|\mathbf{H}_k \mathbf{g}_k\| \leq L \|\mathbf{g}_k\|$. However, (27) is also more generally applicable. For example, when $d = 1$, (27) holds with $\cos^2 \theta = 1$, so long as $\mathbf{H} \neq 0$. Another example, is $f(\mathbf{x}) = \|\mathbf{x}\|^3/6$ where, again, one can show $\cos^2 \theta = 1$.

An immediate consequence of (26) and $\|\mathbf{H}_k \mathbf{g}_k\| \geq |\langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle| / \|\mathbf{g}_k\|$ is that

$$(s_k^{\text{MR}})^2 = \frac{(\langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle)^2}{\|\mathbf{H}_k \mathbf{g}_k\|^4} \leq \frac{\|\mathbf{g}_k\|^4}{(\langle \mathbf{g}_k, \mathbf{H}_k \mathbf{g}_k \rangle)^2} \leq \frac{1}{\mu_0^2}.$$

Therefore by applying Assumption B.9 and (25), we get

$$\|\mathbf{g}(\mathbf{x} + \mathbf{p})\| \leq \left(\frac{L_2}{2\mu_0^2} \|\mathbf{g}\| + \sqrt{1 - \nu_0} \right) \|\mathbf{g}\|. \quad (28)$$

The recursion in the gradient norm, (28), is linear-quadratic in a manner similar to the results obtained for Newton-MR method applied to invex problems; see [63]. It follows directly from (28) that, when the gradient is small enough, i.e.,

$$\|\mathbf{g}\| < G_0 \triangleq 2\mu_0^2 (1 - \sqrt{1 - \nu_0}) / L_2, \quad (29)$$

the gradient norm decreases monotonically. Naturally, once the gradient hits this monotonic phase, every subsequent iterate will also satisfy $\|\mathbf{g}\| \leq G_0$. These properties are all formalized to a rate in Proposition B.11.

Proposition B.11 (MR Scaling Convergence Gradient Norm). *Consider Assumption 3.1, and iterations of the form $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k^{\text{MR}} \mathbf{g}_k$ starting from $\mathbf{x}_0$ for which $\|\mathbf{g}(\mathbf{x}_0)\| < G_0$ and Assumption B.9 holds. For all $k \geq 1$, we have $\|\mathbf{g}_k\| < \|\mathbf{g}_{k-1}\|$ and*

$$\|\mathbf{g}_k\| \leq \frac{G_0}{G_0 - \|\mathbf{g}_0\|} \left(1 - \frac{1 - \sqrt{1 - \nu_0}}{2 - \sqrt{1 - \nu_0}} \right)^k \|\mathbf{g}_0\|.$$

Proof. Since Assumption B.9 holds with $\mathbf{x}_0$ and $\|\mathbf{g}(\mathbf{x}_0)\| < G_0$, (28) implies that $\|\mathbf{g}_k\| \leq \|\mathbf{g}_{k-1}\| < G_0$ for $k \geq 1$. To obtain a quantitative rate, we follow a similar line of reasoning as [52, Theorem 1.2.4]. Let $a_k = \|\mathbf{g}_k\| L_2 / (2\mu_0^2)$ and $q = 1 - \sqrt{1 - \nu_0}$. We have $a_k < q$ and

$$a_{k+1} \leq a_k^2 + \sqrt{1 - \nu_0} a_k = (a_k + \sqrt{1 - \nu_0}) a_k = (1 + a_k - q) a_k.$$

Also, since $a_k - q < 0$ by (29),

$$a_k(1 + a_k - q) = \left(\frac{(1 - (a_k - q))(1 + (a_k - q))}{(1 - (a_k - q))} \right) a_k = \left(\frac{(1 - (a_k - q)^2)}{1 - (a_k - q)} \right) a_k \leq \frac{a_k}{1 + q - a_k},$$

where the inequality follows from $(1 - (a_k - q)^2) \leq 1$. Hence,

$$\frac{q}{a_{k+1}} - 1 \geq (1 + q) \left(\frac{q}{a_k} - 1 \right).$$

Applying this iteratively gives

$$\frac{q}{a_k} - 1 \geq (1 + q)^k \left(\frac{q}{a_0} - 1 \right) = (1 + q)^k \left(\frac{2\mu_0^2(1 - \sqrt{1 - \nu_0})}{L_2 \|\mathbf{g}_0\|} - 1 \right) = (1 + q)^k \left(\frac{G_0}{\|\mathbf{g}_0\|} - 1 \right).$$

Rearranging, we obtain

$$a_k \leq \frac{q \|\mathbf{g}_0\|}{(1 + q)^k (G_0 - \|\mathbf{g}_0\|) + \|\mathbf{g}_0\|}.$$

Finally, by substituting in the definition of a_k and q , we obtain

$$\|\mathbf{g}_k\| \leq \frac{G_0 \|\mathbf{g}_0\|}{(2 - \sqrt{1 - \nu_0})^k (G_0 - \|\mathbf{g}_0\|)} = \frac{G_0}{G_0 - \|\mathbf{g}_0\|} \left(1 - \frac{1 - \sqrt{1 - \nu_0}}{2 - \sqrt{1 - \nu_0}} \right)^k \|\mathbf{g}_0\|.$$

□

Finally, we prove Theorem 3.9 by specializing to the case where the second order sufficient conditions hold.

Proof of Theorem 3.9. It suffices to find a region where the conditions of Proposition B.11 are satisfied, that the iterates will not leave. Similar to the proof of Theorem 3.7, when the second order sufficient conditions hold, there exists a ball $\mathcal{B}_r^*$ such that (9) holds. This in turn implies (21), as well as (26) and (27) with $\mu_0 = \mu$ and $\nu_0 = \mu^2/M^2$, respectively, for any $\mathbf{x} \in \mathcal{B}_r^*$. With these μ_0 and ν_0 in hand, consider G_0 in (29) and define

$$\mathcal{G} = \left\{ \mathbf{x} \mid 0 < \|\mathbf{g}(\mathbf{x})\| \leq \min \left\{ G_0, \frac{r\mu}{M\mu + 1} \right\} \right\}.$$

Suppose $\mathbf{x}_k \in \mathcal{B}_r^* \cap \mathcal{G}$ (such an $\mathbf{x}_k$ exists by the RHS of (21)). From (28), we get $\|\mathbf{g}_{k+1}\| < \|\mathbf{g}_k\|$, which in turn implies $\mathbf{x}_{k+1} \in \mathcal{G}$. Applying the upper bound in (21), $s_k^{\text{MR}} \leq 1/\mu$ and $\mathbf{x}_k \in \mathcal{G}$, gives

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq \|\mathbf{x}_k - \mathbf{x}^*\| + |s_k^{\text{MR}}| \|\mathbf{g}_k\| \leq \left(M + \frac{1}{\mu} \right) \|\mathbf{g}_k\| \leq r.$$

That is, $\mathbf{x}_{k+1} \in \mathcal{B}_r^* \cap \mathcal{G}$. Hence, Assumption B.9 holds for all of the iterates if $\mathbf{x}_0 \in \mathcal{B}_r^* \cap \mathcal{G}$. Proposition B.11 can be applied to obtain the convergence rate. $\square$

Finally, we remark that the iteration $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k^{\text{MR}} \mathbf{g}_k$, as applied in Theorem 3.9, can occur as a special case of Algorithm 2. In particular, if σ is chosen $\sigma < \mu$ then the SPC scaling (in particular, the MR scaling) will always be selected. Moreover, if $\mathbf{x}_0$ is chosen sufficiently close to $\mathbf{x}^*$, (8) will hold; implying that $\alpha = 1$ will be acceptable to the line search. Indeed, (8) will be satisfied for all future iterations by monotonicity of the gradient norm.

B.7 Inexact Hessian

B.7.1 Statement of Results

We now consider the case where the Hessian can only be accessed through an inexact estimate $\tilde{\mathbf{H}}$. Specifically, we demonstrate that under certain conditions on the inexactness tolerance, we can attain similar unit step size and convergence guarantees as in the exact case. For clarity, we defer the proofs of our results to the end of this section. We begin with a condition on the allowable inexactness in the Hessian.

Assumption B.12 (Hessian Error Bound). For a given $\Delta_H > 0$ and $\mathbf{x}$, we can produce $\tilde{\mathbf{H}}$ such that

$$\left| \left\langle \mathbf{g}(\mathbf{x}), (\mathbf{H}(\mathbf{x}) - \tilde{\mathbf{H}}(\mathbf{x})) \mathbf{g}(\mathbf{x}) \right\rangle \right| \leq \Delta_H \|\mathbf{g}(\mathbf{x})\|^2. \quad (30)$$

This condition requires only that the inexactness is bounded along the gradient direction, a weaker requirement than a direct bound on the Hessian error, $\mathbf{H}(\mathbf{x}) - \tilde{\mathbf{H}}(\mathbf{x})$. To ground Assumption B.12, consider an objective which can be written as a finite-sum

$$f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$

This is a formulation which arises often in machine learning as part of the empirical risk minimization framework [51]. In the “big data” regime, where $n \gg 1$, optimization algorithms often use sub-sampling to estimate gradients or Hessians. Suppose we estimate the Hessian at each iteration via

$$\tilde{\mathbf{H}}(\mathbf{x}) = \frac{1}{|\mathcal{I}_H|} \sum_{i \in \mathcal{I}_H} \mathbf{H}_i(\mathbf{x}), \quad (31)$$

where $\mathcal{I}_H \subseteq \{1, \dots, n\}$ is a minibatch of indices uniformly, with replacement. In this case, Assumption B.12 holds with high probability if the sample size is sufficiently large and the individual Hessians, $\mathbf{H}_i$, are uniformly bounded along the gradient direction (see Appendix B.8 for details).

When the estimated Hessian $\tilde{\mathbf{H}}$ replaces $\mathbf{H}$ in Algorithm 1, the curvature tests and resulting scalings (denoted by $\tilde{s}$) depend on $\tilde{\mathbf{H}}$ rather than $\mathbf{H}$. Fortunately, many of the key properties from the deterministic case still hold for inexact Hessian. Indeed, SPC steps satisfy $\sigma \|\mathbf{g}\|^2 < \langle \mathbf{g}, \tilde{\mathbf{H}} \mathbf{g} \rangle$, LPC

steps satisfy $0 \leq \langle \mathbf{g}, \tilde{\mathbf{H}}\mathbf{g} \rangle \leq \sigma \|\mathbf{g}\|^2$ and NC steps satisfy $\langle \mathbf{g}, \tilde{\mathbf{H}}\mathbf{g} \rangle < 0$. The second order descent condition (3) from Proposition 2.3 continues to hold with respect to $\tilde{\mathbf{H}}$, that is, for $\tilde{\mathbf{p}} = -\tilde{s}\mathbf{g}$

$$\langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle \leq 0 \quad (32)$$

The upper bounds from Proposition 2.2 also continue to hold, as they arise due to the curvature test. The search direction, $\tilde{\mathbf{p}}$, remains collinear to the gradient, so that $\langle \mathbf{g}, \tilde{\mathbf{p}} \rangle = -\|\mathbf{g}\| \|\tilde{\mathbf{p}}\|$ and Assumption 3.1 can be applied along $\tilde{\mathbf{p}}$ to ensure that (11) holds. The Armijo condition for the line search is given by

$$f(\mathbf{x} + \alpha\tilde{\mathbf{p}}) \leq f(\mathbf{x}) + \alpha\rho \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle, \quad (33)$$

for $\rho \in (0, 1/2)$. With these ideas in hand we can derive results reminiscent of Section 3.1 for inexact case. Beginning with a unit step size guarantee that applies when Δ_H is sufficiently small.

Proposition B.13. *Consider Assumption 3.1 and Assumption B.12 with $\Delta_H \leq 2 \min \{ \sigma(1/2 - \rho), (1 - \rho)/s_{\max}^{NC} \}$. If*

$$\|\mathbf{g}\| \leq 6 \min \left\{ \frac{\sigma^2 \left(\frac{1}{2} - \rho - \frac{\Delta_H}{2\sigma} \right)}{L_2}, \frac{\left(1 - \rho - \frac{s_{\max}^{NC} \Delta_H}{2} \right)}{L_2 (s_{\max}^{NC})^2} \right\},$$

then the Armijo condition (33) is satisfied with $\alpha = 1$.

Next we consider a global convergence guarantee for the inexact case. Much like the analysis in the exact case, this requires an additional smoothness condition, however this time the condition is based on the inexact Hessian.

Assumption B.14 (Inexact Hessian-gradient Smoothness). *There exists $0 \leq \tilde{L}_1 < \infty$ such that for all $\mathbf{x} \in \mathbb{R}^d$, we have $\|\tilde{\mathbf{H}}(\mathbf{x})\mathbf{g}(\mathbf{x})\| \leq \tilde{L}_1 \|\mathbf{g}(\mathbf{x})\|$.*

Much like the exact Hessian case, Assumption B.14 relaxes the typical Lipschitz gradient assumption used for global convergence in gradient descent. Returning to the sub-sampled Hessian, (31), Assumption B.14 clearly holds if the individual Hessians, $\mathbf{H}_i$, are uniformly bounded along the gradient direction.

Much like the the exact Hessian case (compare with Lemma B.4), under Assumption B.14 we have a lower bound on the scaling magnitudes

$$\frac{\sigma}{\tilde{L}_1^2} \leq s^{\text{MR}} \leq s^{\text{GM}} \leq s^{\text{CG}}. \quad (34)$$

We now state the global convergence result.

Proposition B.15. *Consider Assumption 3.1, Assumption B.12 for any $\Delta_H \geq 0$, and Assumption B.14. Suppose f is lower bounded. For any $0 < \varepsilon_g < 1$, after at most $K \in \mathcal{O}(\varepsilon_g^{-2})$ iterations of Algorithm 2, with $\mathbf{H}$ replaced with $\tilde{\mathbf{H}}$, we have $\|\mathbf{g}_k\| \leq \varepsilon_g$ for some $0 \leq k \leq K$.*

Much like Proposition 3.6, the rate obtained in Proposition B.15 matches the worst case rate for gradient descent on a function with Lipschitz smooth gradients. Interestingly, unlike Proposition B.13, Proposition B.15 imposes no specific requirement on Δ_H . This is because small step sizes (i.e., $\alpha < 1$) can mitigate Hessian noise. Imposing a stronger condition on Δ_H cannot offer an improvement in the overall worst case rate as the inexact rate already matches that of the exact case. However, in Remark B.17, we discuss how bounding Δ_H can offer an improvement in the per-iteration decrease for certain step types.

Finally, we remark that when Assumption B.14 holds only with high probability, the conclusions of Proposition B.13 and Proposition B.15 may also hold only with high probability. A specific issue arises for Proposition B.15, as (30) must hold at *each iteration*. However, by ensuring the probability of (30) failing is sufficiently small, sufficient decrease over any finite span of iterations can be guaranteed with arbitrarily high probability.

B.7.2 Proof of Results

Proof of Proposition B.13. We begin by adding and subtracting $\langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle$ to the upper bound in (11) and applying Assumption B.12

$$\begin{aligned} f(\mathbf{x} + \alpha\mathbf{p}) - f(\mathbf{x}) &\leq \alpha \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \frac{\alpha^2}{2} \langle \tilde{\mathbf{p}}, (\mathbf{H} - \tilde{\mathbf{H}})\tilde{\mathbf{p}} \rangle + \frac{\alpha^2}{2} \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3 \\ &\leq \alpha \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \frac{\alpha^2\Delta_H}{2} \|\tilde{\mathbf{p}}\|^2 + \frac{\alpha^2}{2} \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3. \end{aligned} \quad (35)$$

The upper bound in (35) now involves the curvature of the inexact Hessian, $\langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle$, which we can control with the curvature tests and second order descent condition. Starting with non-negative curvature case (encompassing SPC and LPC scalings) with $\langle \mathbf{g}, \tilde{\mathbf{H}}\mathbf{g} \rangle \geq 0$. Taking (35), subtracting $\rho\alpha \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle$ and applying $\alpha \leq 1$ and (32) we obtain

$$\begin{aligned} f(\mathbf{x} + \alpha\mathbf{p}) - f(\mathbf{x}) - \rho\alpha \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle &\leq \alpha \left(\frac{1}{2} - \rho \right) \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \frac{\alpha}{2} \left(\langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle \right) \\ &\quad + \frac{\alpha^2}{2} \Delta_H \|\tilde{\mathbf{p}}\|^2 + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3 \\ &\leq -\alpha \left(\frac{1}{2} - \rho \right) \|\mathbf{g}\| \|\tilde{\mathbf{p}}\| + \frac{\alpha^2}{2} \Delta_H \|\tilde{\mathbf{p}}\|^2 + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3 \\ &= \left(-\left(\frac{1}{2} - \rho \right) + \frac{\alpha\tilde{s}\Delta_H}{2} + \frac{L_2\alpha^2\tilde{s}}{6} \|\tilde{\mathbf{p}}\| \right) \alpha \|\tilde{\mathbf{p}}\| \|\mathbf{g}\| \\ &\leq \left(-\left(\frac{1}{2} - \rho \right) + \frac{\alpha\Delta_H}{2\sigma} + \frac{L_2\alpha^2}{6\sigma} \|\tilde{\mathbf{p}}\| \right) \alpha \|\tilde{\mathbf{p}}\| \|\mathbf{g}\|. \end{aligned} \quad (36)$$

In the final line we apply $\tilde{s} \leq 1/\sigma$, which holds for both the LPC and SPC scalings. The line search criteria (33) is clearly satisfied if the upper bound in (36) is negative. That is,

$$\|\tilde{\mathbf{p}}\| \leq \frac{6\sigma}{L_2\alpha^2} \left(\left(\frac{1}{2} - \rho \right) - \frac{\alpha\Delta_H}{2\sigma} \right).$$

This bound is nontrivial if

$$\Delta_H \leq \frac{2\sigma}{\alpha} \left(\frac{1}{2} - \rho \right).$$

Now we consider the negative curvature case, where $\langle \mathbf{g}, \tilde{\mathbf{H}}\mathbf{g} \rangle < 0$. Applying the negative curvature condition to (35) and subtracting $\alpha\rho \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle$

$$\begin{aligned} f(\mathbf{x} + \alpha\mathbf{p}) - f(\mathbf{x}) - \alpha\rho \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle &\leq \alpha(1 - \rho) \langle \mathbf{g}, \tilde{\mathbf{p}} \rangle + \frac{\alpha^2\Delta_H}{2} \|\tilde{\mathbf{p}}\|^2 + \frac{\alpha^2}{2} \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3 \\ &\leq -\alpha(1 - \rho) \|\mathbf{g}\| \|\tilde{\mathbf{p}}\| + \frac{\alpha^2\Delta_H}{2} \|\tilde{\mathbf{p}}\|^2 + \frac{L_2\alpha^3}{6} \|\tilde{\mathbf{p}}\|^3 \\ &= \left(-(1 - \rho) + \frac{\alpha\tilde{s}^{\text{NC}}}{2} \Delta_H + \frac{L_2\alpha^2\tilde{s}^{\text{NC}}}{6} \|\tilde{\mathbf{p}}\| \right) \alpha \|\mathbf{g}\| \|\tilde{\mathbf{p}}\| \\ &\leq \left(-(1 - \rho) + \frac{\alpha s_{\max}^{\text{NC}}}{2} \Delta_H + \frac{L_2\alpha^2 s_{\max}^{\text{NC}}}{6} \|\tilde{\mathbf{p}}\| \right) \alpha \|\mathbf{g}\| \|\tilde{\mathbf{p}}\|, \end{aligned} \quad (37)$$

where in the final line we we apply $\tilde{s}^{\text{NC}} \leq s_{\max}^{\text{NC}}$. Again, if the upper bound in (37) is negative (33) is clearly satisfied. In particular, (37) is negative if

$$\|\tilde{\mathbf{p}}\| \leq \frac{6}{L_2 s_{\max}^{\text{NC}} \alpha^2} \left(1 - \rho - \frac{\alpha s_{\max}^{\text{NC}} \Delta_H}{2} \right).$$

This bound is nontrivial if

$$\Delta_H \leq \frac{2(1-\rho)}{\alpha s_{\max}^{\text{NC}}}.$$

The result follows from combining the two tolerances on Δ_H , setting $\alpha = 1$ and using the fact that $\|\tilde{\mathbf{p}}\| \leq \|\mathbf{g}\|/\sigma$ in the non-negative curvature case and $\|\tilde{\mathbf{p}}\| \leq s_{\max}^{\text{NC}} \|\mathbf{g}\|$ in the negative curvature case. $\square$

We now move on to a proof of Proposition B.15. First we require a technical lemma.

Lemma B.16. *For $A \geq 0$, $x \geq 0$ and $B > 0$ we have*

$$-A + \sqrt{A^2 + Bx} \geq \frac{B}{A + \sqrt{A^2 + B}} \min\{x, 1\}.$$

Proof. The $x = 0$ case is trivial, so take $x > 0$. Multiplying the numerator and denominator by a conjugate

$$\begin{aligned} -A + \sqrt{A^2 + Bx} &= \frac{(-A + \sqrt{A^2 + Bx})(A + \sqrt{A^2 + Bx})}{A + \sqrt{A^2 + Bx}} \\ &= \frac{-A^2 + A^2 + Bx}{A + \sqrt{A^2 + Bx}} \\ &= \frac{Bx}{A + \sqrt{A^2 + Bx}}. \end{aligned}$$

Now consider two cases. If $x < 1$ then $A + \sqrt{A^2 + Bx} < A + \sqrt{A^2 + B}$ so that

$$-A + \sqrt{A^2 + Bx} = \frac{Bx}{A + \sqrt{A^2 + Bx}} \geq \frac{B}{A + \sqrt{A^2 + B}} x.$$

On the other hand, if $x \geq 1$, we rearrange to obtain

$$-A + \sqrt{A^2 + Bx} = \frac{Bx}{A + \sqrt{A^2 + Bx}} = \frac{B}{A/x + \sqrt{A^2/x^2 + B/x}}.$$

The denominator of this expression is decreasing in x , indeed, we clearly have $A/x + \sqrt{A^2/x^2 + B/x} \leq A + \sqrt{A^2 + B}$ so that

$$-A + \sqrt{A^2 + Bx} \geq \frac{B}{A + \sqrt{A^2 + B}}.$$

$\square$

Proof of Proposition B.15. We begin by re-examining (36) and (37) from the proof of Proposition B.13, to demonstrate that there is a positive step size for which (33) certainly holds. We assume that Algorithm 2 has not terminated, i.e., $\|\mathbf{g}\| > \varepsilon_g$. Starting with the negative curvature case, consider the upper bound in (37). The line search condition (33) is certainly satisfied if

$$C_2 \alpha^2 + C_1 \alpha + C_0 \leq 0,$$

where

$$C_2 \triangleq \frac{L_2 s_{\max}^{\text{NC}} \|\tilde{\mathbf{p}}\|}{6}, \quad C_1 \triangleq \frac{s_{\max}^{\text{NC}} \Delta_H}{2}, \quad C_0 \triangleq -(1 - \rho).$$

This quadratic has two real roots, one positive and one negative. The positive root is given by

$$\begin{aligned} \alpha_1^* &= -\frac{C_1}{2C_2} + \sqrt{\frac{C_1^2}{4C_2^2} - \frac{C_0}{C_2}} \\ &= -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{6(1-\rho)}{L_2 s_{\max}^{\text{NC}} \|\tilde{\mathbf{p}}\|}}, \end{aligned}$$

which implies that the largest step size for which (33) is satisfied also must satisfy $\alpha \geq \alpha_1^*$. Therefore, when (33) is satisfied, we have

$$\begin{aligned}
f(\mathbf{x} + \alpha \tilde{\mathbf{p}}) - f(\mathbf{x}) &\leq -\alpha \rho \|\tilde{\mathbf{p}}\| \|\mathbf{g}\| \\
&\leq -\rho \left(-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6(1-\rho)\|\tilde{\mathbf{p}}\|}{L_2 s_{\max}^{\text{NC}}}} \right) \|\mathbf{g}\| \\
&< -\rho \left(-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6(1-\rho)s_{\min}^{\text{NC}}\varepsilon_g}{L_2 s_{\max}^{\text{NC}}}} \right) \varepsilon_g \\
&\leq -\rho \left(\frac{6(1-\rho)s_{\min}^{\text{NC}}/(L_2 s_{\max}^{\text{NC}})}{\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6(1-\rho)s_{\min}^{\text{NC}}}{L_2 s_{\max}^{\text{NC}}}}} \right) \varepsilon_g^2 \\
&= -\tilde{c}^{\text{NC}} \varepsilon_g^2.
\end{aligned}$$

On the third line we apply $\|\tilde{\mathbf{p}}\| \geq s_{\min}^{\text{NC}} \|\mathbf{g}\|$ and $\|\mathbf{g}\| > \varepsilon_g$, whereas on the fourth line we apply Lemma B.16 and $\varepsilon_g < 1$. In the final line we define

$$\tilde{c}^{\text{NC}} \triangleq \rho \frac{12(1-\rho)s_{\min}^{\text{NC}}/s_{\max}^{\text{NC}}}{3\Delta_H + \sqrt{9\Delta_H^2 + 24(1-\rho)L_2 s_{\min}^{\text{NC}}/s_{\max}^{\text{NC}}}}.$$

Next we come to the positive curvature case. Considering (36) we see that the line search condition (33) is satisfied if

$$D_2 \alpha^2 + D_1 \alpha + D_0 \leq 0,$$

where

$$D_2 \triangleq \frac{L_2 \|\tilde{\mathbf{p}}\|}{6\sigma}, \quad D_1 \triangleq \frac{\Delta_H}{2\sigma}, \quad D_0 \triangleq -\left(\frac{1}{2} - \rho\right).$$

Again this quadratic has a positive and negative root. The positive root is given by

$$\begin{aligned}
\alpha_2^* &\triangleq -\frac{D_1}{2D_2} + \sqrt{\frac{D_1^2}{4D_2^2} - \frac{D_0}{D_2}} \\
&= -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{6\sigma(1/2 - \rho)}{L_2 \|\tilde{\mathbf{p}}\|}}.
\end{aligned}$$

Therefore, the largest step size for which (33) holds must satisfy $\alpha \geq \min\{1, \alpha_2^*\}$. We now isolate the LPC and SPC cases. For the LPC case, by combining the line search criteria (33) and $\alpha \geq \min\{1, \alpha_2^*\}$ we obtain

$$\begin{aligned}
f(\mathbf{x} + \alpha \tilde{\mathbf{p}}) - f(\mathbf{x}) &\leq -\rho \alpha \|\mathbf{g}\| \|\tilde{\mathbf{p}}\| \\
&\leq -\rho \min\{\alpha_2^* \|\tilde{\mathbf{p}}\|, \|\tilde{\mathbf{p}}\|\} \|\mathbf{g}\| \\
&= -\rho \min\left\{-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma(1/2 - \rho)\|\tilde{\mathbf{p}}\|}{L_2}}, \|\tilde{\mathbf{p}}\|\right\} \|\mathbf{g}\| \\
&< -\rho \min\left\{-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma(1/2 - \rho)s_{\min}^{\text{LPC}}\varepsilon_g}{L_2}}, s_{\min}^{\text{LPC}}\varepsilon_g\right\} \varepsilon_g \\
&\leq -\rho \min\left\{\frac{6\sigma(1/2 - \rho)s_{\min}^{\text{LPC}}/L_2}{\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma(1/2 - \rho)s_{\min}^{\text{LPC}}}{L_2}}}, s_{\min}^{\text{LPC}}\right\} \varepsilon_g^2 \\
&= -\tilde{c}^{\text{LPC}} \varepsilon_g^2.
\end{aligned}$$

On the fourth line we applied $\|\tilde{\mathbf{p}}\| \geq s_{\min}^{\text{LPC}} \|\mathbf{g}\|$ and $\|\mathbf{g}\| > \varepsilon_g$, while in the fifth line we applied Lemma B.16 and $\varepsilon_g < 1$. In the final line we define

$$\tilde{c}^{\text{LPC}} \triangleq \rho \min \left\{ \frac{12\sigma(1/2 - \rho)s_{\min}^{\text{LPC}}}{3\Delta_H + \sqrt{9\Delta_H^2 + 24\sigma(1/2 - \rho)L_2s_{\min}^{\text{LPC}}}}, s_{\min}^{\text{LPC}} \right\}.$$

Finally, for the SPC case the line search condition (33) and $\alpha \geq \min\{1, \alpha_2^*\}$ yields

$$\begin{aligned} f(\mathbf{x} + \alpha\tilde{\mathbf{p}}) - f(\mathbf{x}) &\leq -\rho\alpha \|\mathbf{g}\| \|\tilde{\mathbf{p}}\| \\ &\leq -\rho \min\{\alpha_2^* \|\tilde{\mathbf{p}}\|, \|\tilde{\mathbf{p}}\|\} \|\mathbf{g}\| \\ &= -\rho \min \left\{ -\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma(1/2 - \rho)\|\tilde{\mathbf{p}}\|}{L_2}}, \|\tilde{\mathbf{p}}\| \right\} \|\mathbf{g}\| \\ &< -\rho \min \left\{ -\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma^2(1/2 - \rho)\varepsilon_g}{L_2\tilde{L}_1^2}}, \frac{\sigma}{\tilde{L}_1^2}\varepsilon_g \right\} \varepsilon_g \\ &\leq -\rho \min \left\{ \frac{6\sigma^2(1/2 - \rho)/(L_2\tilde{L}_1^2)}{\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma^2(1/2 - \rho)}{L_2\tilde{L}_1^2}}}, \frac{\sigma}{\tilde{L}_1^2} \right\} \varepsilon_g^2 \\ &= -\tilde{c}^{\text{SPC}} \varepsilon_g^2. \end{aligned}$$

We apply $\|\tilde{\mathbf{p}}\| \geq (\sigma/\tilde{L}_1^2) \|\mathbf{g}\|$, which follows from (34), and $\|\mathbf{g}\| > \varepsilon_g$ in the fourth line, while in the fifth line we apply Lemma B.16. In the final line we define

$$\tilde{c}^{\text{SPC}} \triangleq \rho \min \left\{ \frac{12\sigma^2(1/2 - \rho)/\tilde{L}_1^2}{3\Delta_H + \sqrt{9\Delta_H^2 + 24\sigma^2(1/2 - \rho)L_2/\tilde{L}_1^2}}, \frac{\sigma}{\tilde{L}_1^2} \right\}.$$

The proof now proceeds similarly to the deterministic case. Let

$$K \triangleq \left\lceil \frac{f(\mathbf{x}_0) - f^*}{\min\{\tilde{c}^{\text{NC}}, \tilde{c}^{\text{LPC}}, \tilde{c}^{\text{SPC}}\} \varepsilon_g^2} \right\rceil.$$

Suppose that $\|\mathbf{g}_k\| \leq \varepsilon_g$ fails to hold for $k = 0, \dots, K-1$. We divide the iterations $\mathcal{K} = \{0, \dots, K-1\}$ into a disjoint union $\mathcal{K} = \mathcal{K}_{\text{NC}} \cup \mathcal{K}_{\text{LPC}} \cup \mathcal{K}_{\text{SPC}}$ where

$$\begin{aligned} \mathcal{K}_{\text{NC}} &= \left\{ k \in \mathcal{K} \mid \langle \mathbf{g}_k, \tilde{\mathbf{H}}_k \mathbf{g}_k \rangle < 0 \right\} \\ \mathcal{K}_{\text{LPC}} &= \left\{ k \in \mathcal{K} \mid 0 \leq \langle \mathbf{g}_k, \tilde{\mathbf{H}}_k \mathbf{g}_k \rangle \leq \sigma \|\mathbf{g}_k\|^2 \right\} \\ \mathcal{K}_{\text{SPC}} &= \left\{ k \in \mathcal{K} \mid \langle \mathbf{g}_k, \tilde{\mathbf{H}}_k \mathbf{g}_k \rangle > \sigma \|\mathbf{g}_k\|^2 \right\}. \end{aligned}$$

Applying a telescoping sum and the lower bounds on function decrease in each case, we have

$$\begin{aligned} f(\mathbf{x}_0) - f(\mathbf{x}_K) &= \sum_{i=0}^{K-1} f(\mathbf{x}_i) - f(\mathbf{x}_{i+1}) \\ &= \sum_{k \in \mathcal{K}_{\text{NC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) + \sum_{k \in \mathcal{K}_{\text{LPC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) + \sum_{k \in \mathcal{K}_{\text{SPC}}} f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \\ &> |\mathcal{K}_{\text{NC}}| \tilde{c}^{\text{NC}} \varepsilon_g^2 + |\mathcal{K}_{\text{LPC}}| \tilde{c}^{\text{LPC}} \varepsilon_g^2 + |\mathcal{K}_{\text{SPC}}| \tilde{c}^{\text{SPC}} \varepsilon_g^2 \\ &\geq (|\mathcal{K}_{\text{NC}}| + |\mathcal{K}_{\text{LPC}}| + |\mathcal{K}_{\text{SPC}}|) \min\{\tilde{c}^{\text{NC}}, \tilde{c}^{\text{LPC}}, \tilde{c}^{\text{SPC}}\} \varepsilon_g^2 \\ &= K \min\{\tilde{c}^{\text{NC}}, \tilde{c}^{\text{LPC}}, \tilde{c}^{\text{SPC}}\} \varepsilon_g^2 \\ &\geq f(\mathbf{x}_0) - f^*, \end{aligned}$$

which implies $f(\mathbf{x}_K) < f^*$, a contradiction. $\square$

Remark B.17 (Remark on Proposition B.15). Notably, Proposition B.15 does not require an explicit bound on the Hessian error, Δ_H . However, as we now demonstrate, the per iteration dependence on ε_g can be improved (relative to the worst case) for SPC steps if we impose a bound on Δ_H . In particular, we stipulate that, for $\theta \in [0, 1)$, the Hessian error tolerance satisfies

$$\Delta_H \leq 2\theta\sigma(1/2 - \rho). \quad (38)$$

Consider then a “large $\tilde{\mathbf{p}}$ ” setting defined by

$$\|\tilde{\mathbf{p}}\| \geq \frac{6\sigma(1/2 - \rho) - 3\Delta_H}{L_2} = \frac{6\sigma(1/2 - \rho)}{L_2} - \frac{3\Delta_H}{L_2}.$$

Note that the lower bound on $\tilde{\mathbf{p}}$ is nontrivial because of the bound in (38). Rearranging and subbing to the step size acceptance tolerance for the SPC case, α_2^* , we have

$$\begin{aligned} \alpha_2^* &= -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{6\sigma(1/2 - \rho)}{L_2 \|\tilde{\mathbf{p}}\|}} \\ &\leq -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{\|\tilde{\mathbf{p}}\| + \frac{3\Delta_H}{L_2}}{\|\tilde{\mathbf{p}}\|}} \\ &= -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{1 + \frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{3\Delta_H}{L_2 \|\tilde{\mathbf{p}}\|}} \\ &= -\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\left(1 + \frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|}\right)^2} \\ &= 1. \end{aligned}$$

Therefore, for large $\tilde{\mathbf{p}}$ the largest step size that satisfies (33), must also satisfy $\alpha \geq \alpha_2^*$. Additionally, applying (38) to the lower bound on $\|\tilde{\mathbf{p}}\|$ we obtain

$$\|\tilde{\mathbf{p}}\| \geq \frac{6\sigma(1/2 - \rho) - 3\Delta_H}{L_2} \geq \frac{6\sigma(1 - \theta)(1/2 - \rho)}{L_2}.$$

By combining the line search condition (33), $\alpha \geq \alpha_2^*$, (32), the SPC curvature test and the lower bound on $\|\tilde{\mathbf{p}}\|$ we have

$$\begin{aligned} f(\mathbf{x} + \alpha\tilde{\mathbf{p}}) - f(\mathbf{x}) &\leq \alpha\rho \langle \tilde{\mathbf{p}}, \mathbf{g} \rangle \\ &\leq -\alpha_2^*\rho \langle \tilde{\mathbf{p}}, \tilde{\mathbf{H}}\tilde{\mathbf{p}} \rangle \\ &\leq -\rho \left(-\frac{3\Delta_H}{2L_2 \|\tilde{\mathbf{p}}\|} + \sqrt{\frac{9\Delta_H^2}{4L_2^2 \|\tilde{\mathbf{p}}\|^2} + \frac{6\sigma(1/2 - \rho)}{L_2 \|\tilde{\mathbf{p}}\|}} \right) \sigma \|\tilde{\mathbf{p}}\|^2 \\ &\leq -\sigma\rho \left(-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{6\sigma(1/2 - \rho) \|\tilde{\mathbf{p}}\|}{L_2}} \right) \|\tilde{\mathbf{p}}\| \\ &\leq -\sigma\rho \left(-\frac{3\Delta_H}{2L_2} + \sqrt{\frac{9\Delta_H^2}{4L_2^2} + \frac{36\sigma^2(1/2 - \rho)^2(1 - \theta)}{L_2^2}} \right) \left(\frac{6\sigma(1 - \theta)(1/2 - \rho)}{L_2} \right) \\ &= -\sigma\rho \left(\frac{72\sigma^2(1/2 - \rho)^2(1 - \theta)/L_2}{3\Delta_H + \sqrt{9\Delta_H^2 + 144\sigma^2(1/2 - \rho)^2(1 - \theta)}} \right) \left(\frac{6\sigma(1 - \theta)(1/2 - \rho)}{L_2} \right) \\ &= -\frac{\sigma\rho}{L_2^2} \left(\frac{432\sigma^3(1/2 - \rho)^3(1 - \theta)^2}{3\Delta_H + \sqrt{9\Delta_H^2 + 144\sigma^2(1/2 - \rho)^2(1 - \theta)}} \right). \quad (39) \end{aligned}$$

This bound suggests that, analogously to the “large $\mathbf{p}$ ” case for the exact Hessian, a better dependence on ε_g is obtained when $\tilde{\mathbf{p}}$ is large and Δ_H is controlled. In fact, when $\theta = 0$ we have $\Delta_H = 0$ and (39) matches the “large $\mathbf{p}$ ” exact case in Lemma B.2. In conclusion, despite the worst case headline rate, in some cases a better dependence on ε_g may be obtained for certain steps if Δ_H is controlled.

B.8 Sub-sampled Hessian Error Bound

Lemma B.18 (Hessian Sub-sampling). *Consider the finite-sum objective*

$$f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$

Suppose that there exists $0 \leq L_1^{\max} < \infty$ such that for all $\mathbf{x} \in \mathbb{R}^d$ we have $\max_{i=1, \dots, n} \|\mathbf{H}_i \mathbf{g}\| \leq L_1^{\max} \|\mathbf{g}\|$. Let $\delta \in (0, 1)$ and $\tilde{\mathbf{H}}$ be as in (31). Then for any $\Delta_H \geq 0$ and $\mathbf{x} \in \mathbb{R}^d$ we have

$$\mathbb{P} \left(\left\| (\tilde{\mathbf{H}} - \mathbf{H}) \mathbf{g} \right\| \leq \Delta_H \|\mathbf{g}\| \right) \geq 1 - \delta,$$

if

$$|\mathcal{I}_H| \geq \frac{(L_1^{\max})^2 \left(1 + \sqrt{8 \log(1/\delta)}\right)^2}{\Delta_H^2}.$$

Proof. The result proceeds similarly to Roosta and Mahoney [64, Lemma 3]. First we write $\mathbf{H}\mathbf{g} = \mathbf{A}\mathbf{B}$ where

$$\mathbf{A} = [\mathbf{H}_1 \mathbf{g}, \dots, \mathbf{H}_n \mathbf{g}] \in \mathbb{R}^{d \times n}, \quad \mathbf{B} = [1/n, \dots, 1/n]^\top \in \mathbb{R}^n.$$

We would like to relate this matrix product to the sub-sampled alternative. To do this, take the minibatch, $\mathcal{I}_H$, and form $\tilde{\mathbf{A}} \in \mathbb{R}^{d \times |\mathcal{I}_H|}$, using the columns of $\mathbf{A}$ corresponding to $\mathcal{I}_H$, rescaled by $\sqrt{n/|\mathcal{I}_H|}$. Similarly, form $\tilde{\mathbf{B}} \in \mathbb{R}^{|\mathcal{I}_H|}$, using the rows of $\mathbf{B}$ corresponding to $\mathcal{I}_H$, rescaled by $\sqrt{n/|\mathcal{I}_H|}$. Clearly,

$$\tilde{\mathbf{A}}\tilde{\mathbf{B}} = \frac{n}{|\mathcal{I}_H|} \sum_{i \in \mathcal{I}_H} \frac{1}{n} \mathbf{H}_i \mathbf{g} = \tilde{\mathbf{H}}\mathbf{g}.$$

Applying Drineas et al. [22, Lemma 11] allows us to relate $\mathbf{A}\mathbf{B}$ to $\tilde{\mathbf{A}}\tilde{\mathbf{B}}$. In particular, with probability at least $1 - \delta$, we have

$$\begin{aligned} \left\| \mathbf{H}\mathbf{g} - \tilde{\mathbf{H}}\mathbf{g} \right\| &= \left\| \mathbf{A}\mathbf{B} - \tilde{\mathbf{A}}\tilde{\mathbf{B}} \right\| \leq \sqrt{\frac{n}{|\mathcal{I}_H|} \sum_{i=1}^n \|\mathbf{A}_i\|^2 |\mathbf{B}_i|^2} + \frac{n}{\sqrt{|\mathcal{I}_H|}} \sqrt{8 \log(1/\delta)} \max_{i=1, \dots, n} \|\mathbf{A}_i\| |\mathbf{B}_i| \\ &\leq \sqrt{\frac{1}{n|\mathcal{I}_H|} \sum_{i=1}^n \|\mathbf{H}_i \mathbf{g}\|^2} + \frac{n}{\sqrt{|\mathcal{I}_H|}} \sqrt{8 \log(1/\delta)} \max_{i=1, \dots, n} \|\mathbf{H}_i \mathbf{g}\| / n \\ &\leq \sqrt{\frac{1}{|\mathcal{I}_H|} (L_1^{\max})^2 \|\mathbf{g}\|^2} + \sqrt{\frac{8 \log(1/\delta)}{|\mathcal{I}_H|} L_1^{\max} \|\mathbf{g}\|} \\ &\leq \frac{1}{\sqrt{|\mathcal{I}_H|}} \left(1 + \sqrt{8 \log(1/\delta)}\right) L_1^{\max} \|\mathbf{g}\|. \end{aligned}$$

Finally, we see that

$$|\mathcal{I}_H| \geq \frac{(L_1^{\max})^2 \left(1 + \sqrt{8 \log(1/\delta)}\right)^2}{\Delta_H^2} \implies \frac{1}{\sqrt{|\mathcal{I}_H|}} \left(1 + \sqrt{8 \log(1/\delta)}\right) L_1^{\max} \leq \Delta_H.$$

□

As a result of Lemma B.18 it is clear that Assumption B.12 is satisfied with high probability as, by the Cauchy-Schwarz inequality, we have

$$\left| \left\langle \mathbf{g}, (\mathbf{H} - \tilde{\mathbf{H}}) \mathbf{g} \right\rangle \right| \leq \|\mathbf{g}\| \left\| (\mathbf{H} - \tilde{\mathbf{H}}) \mathbf{g} \right\|.$$

C Miscellaneous Scaling Properties

MR Scaling Tighter Lower Bound Recall from Lemma B.4 that if $\|\mathbf{Hg}\| \leq L_1 \|\mathbf{g}\|$ for some L_1 (see Assumption 3.4) then CG, MR and GM scalings satisfy a lower bound in terms of the gradient norm. Notably, the MR scaling lower bound is weaker by a factor of σ/L_1 , in comparison with the CG and GM. In the following lemma we show that this lower bound can be tightened to match the CG and GM case if the Hessian spectrum is positive and bounded.

Lemma C.1. *If the Hessian satisfies $0 \preceq \mathbf{H} \preceq M\mathbf{I}$ and $\|\mathbf{Hg}\| > 0$ then*

$$s^{\text{MR}} \geq 1/M.$$

Proof. We utilise the fact that s^{MR} is a Rayleigh quotient of $\mathbf{H}^\dagger$. Suppose that $\mathbf{H}$ has ψ_+ nonzero eigenvalues ($\|\mathbf{Hg}\| > 0$ implies $\psi_+ > 0$) given by $0 < \lambda_1 < \lambda_2 \leq \dots \leq \lambda_{\psi_+} \leq M$. The corresponding nonzero eigenvalues of $\mathbf{H}^\dagger$ are given by $1/\lambda_{\psi_+} \leq \dots \leq 1/\lambda_1$. Since $\mathbf{Hg} \in \text{Range}(\mathbf{H})$ and is therefore orthogonal to $\text{Null}(\mathbf{H})$ we have

$$s^{\text{MR}} = \frac{\langle \mathbf{g}, \mathbf{Hg} \rangle}{\|\mathbf{Hg}\|^2} = \frac{\langle \mathbf{Hg}, \mathbf{H}^\dagger \mathbf{Hg} \rangle}{\|\mathbf{Hg}\|^2} \geq \frac{1}{\lambda_{\psi_+}} \frac{\|\mathbf{Hg}\|^2}{\|\mathbf{Hg}\|^2} \geq \frac{1}{M}.$$

□

The requirement that $\|\mathbf{Hg}\| > 0$ is not stringent since the positive curvature test $\langle \mathbf{g}, \mathbf{Hg} \rangle > 0$ implies $\mathbf{Hg} \neq 0$.

D Additional Numerical Results

The code used to run our experiments can be found [here](#).

D.1 Oracle Calls as Complexity Measure

Following the typical convention in the optimization literature, in all our experiments, we plot the objective value against the total number of oracle calls for function, gradient, and Hessian-vector product evaluations, which allows for a fair comparison between methods with a differing per-iteration computational costs. We adopt this approach because the measurement of “wall-clock” time can be heavily dependent on specific implementation details and computational platform. In contrast, counting the number of equivalent function evaluations, as an implementation and system independent unit of complexity is more appropriate and fair. More specifically, upon evaluating the function, computing its gradient is equivalent to one additional function evaluation, and computing a Hessian-vector product requires two additional function evaluations compared to a gradient evaluation [61]. For example, in neural networks, for given data at the input layer, evaluation of network’s output, i.e., function evaluation, involves one forward propagation. The corresponding gradient is computed by performing one additional backward propagation. After computing the gradient, an additional forward followed by a backward propagation give the corresponding Hessian-vector product [9, 27].

D.2 Competitor Algorithms

In this section we give an outline of the algorithms we are comparing against.

Line Search For line search gradient descent, we apply a backtracking procedure (Algorithm 3) based on the Amijo condition (7) with $\mathbf{p} = -\mathbf{g}$. Inspired by [68, Algorithm 2], we consider three “reset schemes” for initializing back tracking line search at each iteration, i.e., setting α^0 in Algorithm 3. In the first procedure, which we term *no reset*, we set $\alpha^0 = \alpha_{k-1}$, where α_{k-1} is the final step size from the previous iteration. Because the final step size is recycled at each iteration, no reset initialization will require significantly less backtracking. However, by the same token, once the step size shrinks it cannot be increased in future iterations, which limits step sizes and adaptivity. In contrast, for *full reset* initialization we fix some α^{init} and set $\alpha^0 = \alpha^{\text{init}}$ at each iteration. This setting will lead to additional backtracks as the line search must begin over at each iteration, however it also allows for a greater degree of adaptivity when compared with no resetting. Finally, in *limited reset*

we set $\alpha^0 = \alpha_{k-1}/\gamma$ where $\gamma \in (0, 1)$ is a growth parameter and α_{k-1} is the final step size from the previous iteration. This setting is an intermediate between full and no resets, as it allows for some growth in the step length from iteration to iteration. For our experiments we set $\gamma = \theta$, $\alpha^{\text{init}} = 1$ and $\alpha_{-1} = 1$. We report the resetting technique utilized in our final results in the additional results for each of our examples (Appendices D.3 to D.5).

PoNo Line Search In addition to the above line search “reset” schemes, we also consider a deterministic version of the Polyak non-monotone line search from [25]. Specifically, we simply replace the minibatch function and gradient with the full function and gradient values. We set the hyperparameters to the “standard” values outlined in [25, Appendix C] and use $f^* = 0$ as our lower bound.

Fixed Step Size and Accelerated Methods Fixed step size GD consists of applying the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \mathbf{g}_k, \quad (40)$$

at each iteration, where α is a step size parameter which must be set. The momentum methods utilize updates of the form

$$\begin{cases} \mathbf{v}_{k+1} = \beta \mathbf{v}_k - \alpha \mathbf{g}_k \\ \mathbf{x}_{k+1} = \begin{cases} \mathbf{x}_k + \mathbf{v}_{k+1} & \text{if Heavy ball.} \\ \mathbf{x}_k - \alpha \mathbf{g}_k + \beta \mathbf{v}_{k+1} & \text{if Nesterov.} \end{cases} \end{cases} \quad (41)$$

where α is a step size parameter and β is a momentum parameter. If no theoretical value is available we set $\beta = 0.9$. For Adam [39] we use standard values of $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. Each of the Adam, fixed step size and momentum methods require a step size parameter to be set. We do this by using either theoretical values (where available) or via a tuning procedure; see Appendices D.3 to D.5 for specifics.

D.3 Multi-class Logistic Regression

Consider a set of data items $\mathcal{D} = \{\mathbf{a}_i, b_i\}_{i=1}^n \subset \mathbb{R}^d \times \{1, \dots, C\}$. Denote the weights of each class as $\mathbf{x}_1, \dots, \mathbf{x}_C$ and define $\mathbf{x} = [\mathbf{x}_1, \dots, \mathbf{x}_{C-1}]$. We are free to take $\mathbf{x}_C = \mathbf{0}$ as class C is identifiable from the weights of the other classes. The objective, f , is given by

$$f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \sum_{c=1}^{C-1} -\mathbf{1}(b_i = c) \log(\text{softmax}(\mathbf{x}_c, \mathbf{a}_i)) + \frac{\lambda}{2} \|\mathbf{x}\|^2, \quad (42)$$

where $\mathbf{1}(\cdot)$ is the indicator function and

$$\text{softmax}(\mathbf{x}_c, \mathbf{a}_i) = \frac{\exp(\langle \mathbf{x}_c, \mathbf{a}_i \rangle)}{\sum_{c=1}^C \exp(\langle \mathbf{x}_c, \mathbf{a}_i \rangle)}.$$

In our experiments a bias parameter is included in the weights for each class. We set the regularization parameter to $\lambda = 10^{-3}$ and initialize by setting $\mathbf{x}_0 = \mathbf{0}$. We run all methods until a maximum of 10^5 oracle calls or until $\|\mathbf{g}_k\| \leq 10^{-4}$.

Note that the spectrum of the Hessian of (42) can be bounded as

$$\lambda \mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq \left(\frac{C-1}{4n} \|\mathbf{A}\|^2 + \lambda \right) \mathbf{I}, \quad (43)$$

which allows us to estimate the strong convexity parameter and Lipschitz constant of (42) as

$$\mu_{\text{approx}} \triangleq \lambda, \quad L_{\text{approx}} \triangleq \frac{C-1}{4n} \|\mathbf{A}\|^2 + \lambda.$$

Parameter Settings For the scaled gradient methods, we set $\sigma = 0$ as (42) is μ -strongly convex (i.e., curvature along the gradient is already lower bounded). For fixed step size gradient descent (40) we use a step size of $\alpha = 1/L_{\text{approx}}$. For both the scaled GD and vanilla GD line search we utilized the standard settings $\theta = 0.5$ and $\rho = 10^{-4}$. The limited reset initialization scheme achieved the best performance for vanilla GD with line search.

For μ -strongly convex, L_g -smooth problems the parameters for the Heavy ball momentum [62] can be set as

$$\alpha = \frac{4}{(\sqrt{L_g} + \sqrt{\mu})^2}, \quad \beta = \left(\frac{\sqrt{L_g} - \sqrt{\mu}}{\sqrt{L_g} + \sqrt{\mu}} \right)^2.$$

While for Nesterov acceleration [52] the step size and momentum parameter can be set as

$$\alpha = \frac{1}{L_g}, \quad \beta = \frac{\sqrt{L_g/\mu} - 1}{\sqrt{L_g/\mu} + 1}.$$

For the momentum methods (41) in our experiments use these settings with μ_{approx} and L_{approx} taking the place of μ and L_g , respectively. The step size for Adam was tuned over the grid $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$.

Additional Results We now present some additional numerical results for the multi-class logistic regression problem. In Figure 5 we give the results from Figure 2 with PoNo line search included. We see that, despite enforcing monotonicity, at each iteration our method outperforms the best iterates of the highly non-monotone PoNo method. Meanwhile, the best iterates of PoNo outperform vanilla GD with line search.

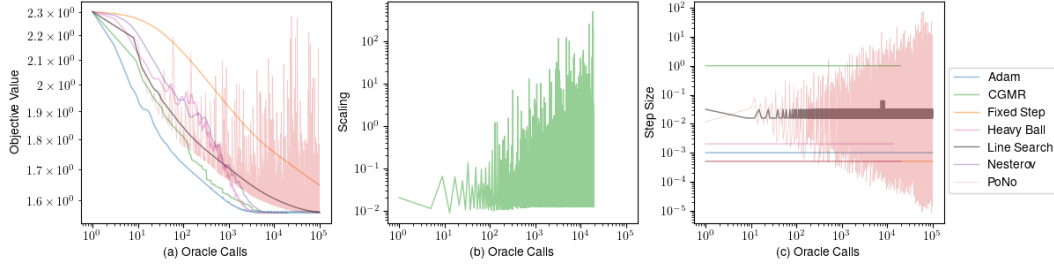


Figure 5: Multi-class logistic regression on CIFAR10 (Figure 2 with PoNo line search included). (a) Objective value. (b) Scaling utilized by the CGMR method. (c) Step size.

In Figure 6 we evaluate the performance of each of the scaled gradient methods. We see that CGMR and MRCG (which differ by whether they start on the CG or MR scaling, respectively) perform quite similarly and handily outperform the other methods.

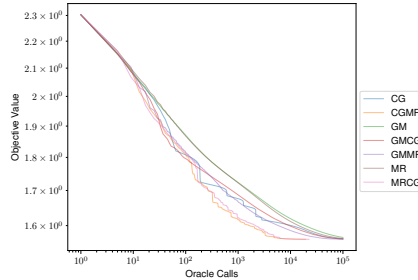


Figure 6: Comparison between scaling method performance for multiclass logistic regression on CIFAR10.

In Figure 7 and Figure 8 we see a breakdown of the performance of the “CG” and “MR” scalings, respectively. Firstly, in panel (c) for each method, we see that the unit step size is accepted by the line search at each iteration, similarly to MRCG. For the MR scaling (Figure 8) we plot the gradient norm in the (a) panel, from which we see that the MR method produces a monotonic decrease in the gradient norm at each iteration with a unit step size; indicating the results from Theorem 3.9 can hold over a wide portion of the optimization landscape. This is despite the line search targeting the

function value, rather than the gradient norm. In panel (b) of Figure 7 and Figure 8 we see that the oscillating behavior is observed in the scaling values. However, comparing with Figure 2, it is clear that this oscillation is over a smaller range of values (particularly for MR scaling) than the alternating MRCG scaling. This indicates that the alternation between MR and CG scalings is key to obtaining large scaling values and, in light of Theorem 3.7, corresponding rapid convergence.

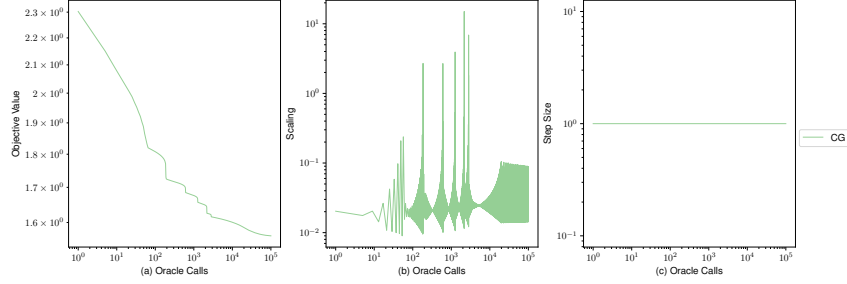


Figure 7: Performance of the CG scaling on the multi-class logistic regression problem on CIFAR10. (a) The objective function. (b) The scaling selected by the CG method. (c) The step size selected by line search.

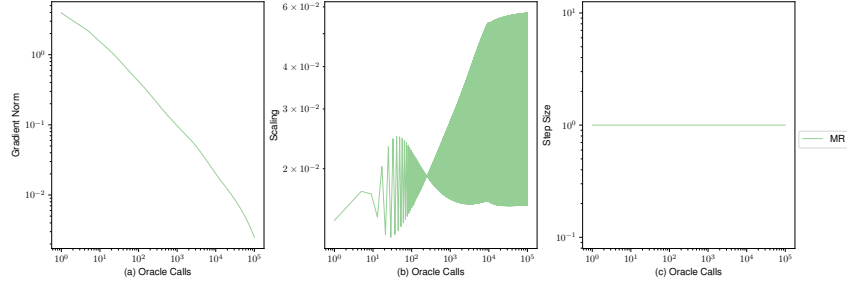


Figure 8: Performance of the MR scaling on the multi-class logistic regression problem on CIFAR10. (a) The gradient norm. (b) The scaling selected by the MR method. (c) The step size selected by line search.

Finally, in Figure 9 we plot the objective value against wall-clock time we see that results largely conform with oracle calls.

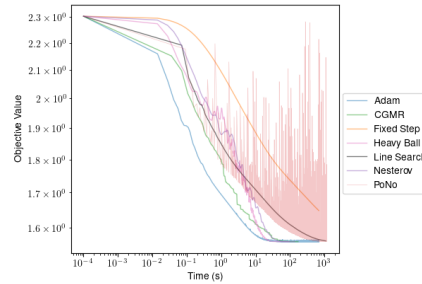


Figure 9: Wall-clock time for multi-class logistic regression on the CIFAR10 dataset. 10^{-4} has been added to all times for plotting purposes.

D.4 MLP

Let $\mathbf{h}(\mathbf{x}; \cdot)$ denote a two layer MLP with 100 hidden units per layer with GeLU [31] activations and C output layers, with parameters $\mathbf{x}$. The objective for our experiment is the function

$$f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \text{CrossEntropy}(\mathbf{h}(\mathbf{x}; \mathbf{a}_i), b_i) + \frac{\lambda}{2} \|\mathbf{x}\|^2, \quad (44)$$

where $\text{CrossEntropy}(\cdot, \cdot)$ is the cross-entropy loss between the predictions of the MLP and the true labels and λ is a ℓ_2 regularization (or weight decay) parameter. In our experiments the regularization parameter was set to $\lambda = 10^{-3}$ and the parameters were initialised with the PyTorch default, that is, the weights for each layer are drawn from $U(-\sqrt{k}, \sqrt{k})$ where $k = 1/(\text{\#num inputs to the layer})$. We run all methods until a maximum of 10^5 oracle calls or until $\|\mathbf{g}_k\| \leq 10^{-4}$.

Parameter Settings For the scaled methods we set $\sigma = 10^{-6}$ and utilize fixed scalings for the LPC and NC case given by $s^{\text{LPC}} = s^{\text{NC}} = 1$. For the scaled and vanilla GD line search methods we utilize the standard values $\theta = 0.5$ and $\rho = 10^{-4}$. For vanilla gradient descent with line search, limited reset initialization achieved the best performance.

Since (44) is nonconvex and there is no closed expression for the gradient Lipschitz constant, we manually tuned the step size parameters for fixed step, momentum methods and Adam. The tuning grid and resulting learning rate parameters are summarized in Table 1.

Method	Grid	Learning Rate
Fixed	$\{10^0, 10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-2}
Adam	$\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$	10^{-5}
HB	$\{10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-3}
NES	$\{10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-3}

Table 1: Tuned learning rates for MLP model on FashionMNIST

Additional Results We report some additional results for the MLP model on FashionMNIST dataset. In Figure 10 we plot the main body results (Figure 3) with PoNo line search included. Again we see that our method outperforms the best iterates of the non-monotone PoNo method. Meanwhile, the best iterates of the PoNo method significantly outperform vanilla line search.

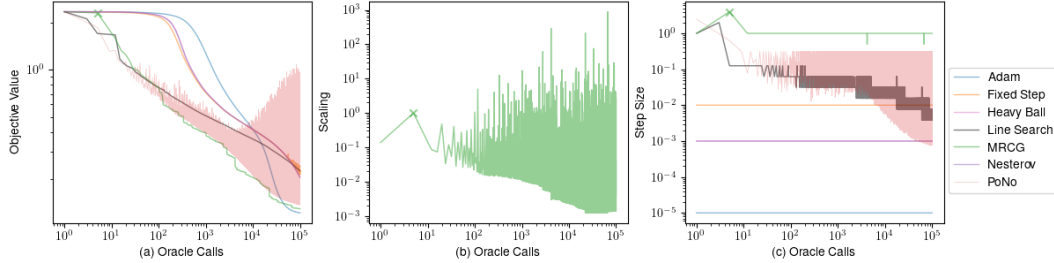


Figure 10: MLP on the FashionMNIST (Figure 3 with PoNo line search included). (a) Objective value (b) Scaling utilized by the MRCG method (c) Step size. Crosses indicate iterations where negative curvature is detected.

In Figure 11 we compare the performance of each of the scalings on the problem. Again, we see that the alternating scalings “CGMR” and “MRCG” perform the best.

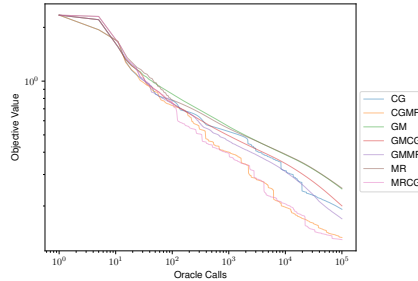


Figure 11: Comparison between scaling methods for MLP on FashionMNIST dataset.

In Figure 12 and Figure 13 we consider the performance of the CG and MR methods, respectively. For the CG and MR scaling we see that the unit step length is accepted at each iteration except for the iteration where NC is detected, consistent with Theorem 3.3. When NC is detected forward tracking line search kicks in and a larger step size is selected. Similarly, to the logistic regression case, in panel (a) of Figure 13 we see that the gradient norm is monotonic, except for iteration where NC is detected, reinforcing the results in Theorem 3.9. No LPC directions are detected for either the MR or CG scaling.

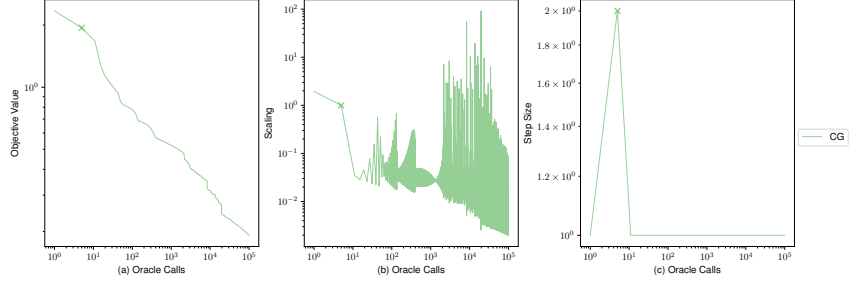


Figure 12: Performance of the CG scaling for the MLP on the FashionMNIST dataset. (a) The objective function. (b) The scaling selected by the CG method. (c) The step size selected by line search. Crosses indicate iterations where negative curvature is detected.

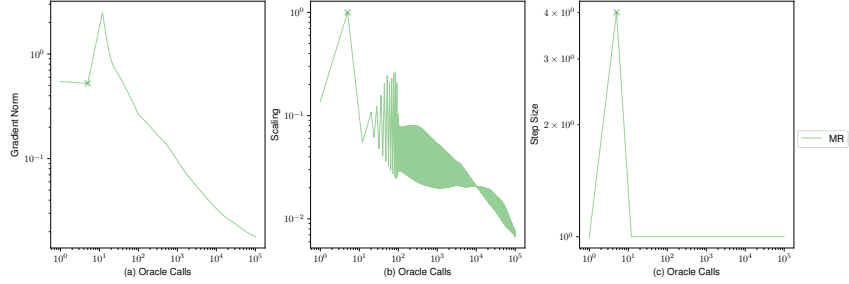


Figure 13: Performance of the MR scaling for the MLP on the FashionMNIST dataset. (a) The gradient norm. (b) The scaling selected by the MR method. (c) The step size selected by line search. Crosses indicate iterations where negative curvature is detected.

In Figure 14 we report our results from Figure 3 in terms of wall-clock time. We see that the results are largely unchanged.

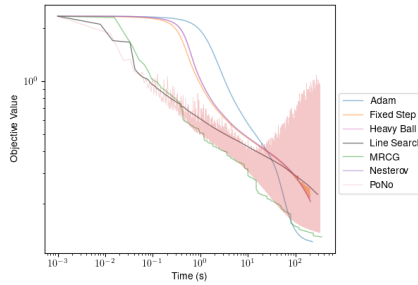


Figure 14: Wall-clock time results for MLP on the FashionMNIST dataset [71]. 10^{-3} has been added to all times for plotting purposes.

D.5 Resnet

Let $\mathbf{h}(\mathbf{x}, \cdot)$ denote a depth 18 ResNet [30] with parameters $\mathbf{x}$. The objective for our experiment is the function

$$f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \text{CrossEntropy}(\mathbf{h}(\mathbf{x}; \mathbf{a}_i), b_i) + \frac{\lambda}{2} \|\mathbf{x}\|^2, \quad (45)$$

where $\text{CrossEntropy}(\cdot, \cdot)$ is the cross-entropy loss between the predictions of the network and the true labels and λ is an ℓ_2 regularization parameter. We specifically consider off-the-shelf ResNet18 implementation from PyTorch⁶ with two modifications. Firstly, we replace all ReLU activations with GeLU [31] to ensure twice differentiability. The second modification is replacing the BatchNorm [34] layers with LayerNorm [4]. This modification has previously been considered in [70]. We utilize the “160px” version of the Imagenette dataset [33], which is available from PyTorch [58]. The Imagenette is a 10 class, subset of the full Imagenet dataset [21]. We process the data by normalizing per channel followed by resizing the images to 32×32 .

In our experiments we set $\lambda = 10^{-5}$ and use the default initialization for the PyTorch ResNet model. All experiments are run until a maximum of 10^4 oracle calls or $\|\mathbf{g}_k\| \leq 10^{-4}$.

Parameter Settings For our scaled gradient method we set $\sigma = 10^{-6}$ and $s^{\text{NC}} = s^{\text{LPC}} = 1$. Our scaled and vanilla GD line searches utilized $\theta = 0.5$, and $\rho = 0$. This choice of ρ was made for numerical stability purposes, as, in contrast to the previous two experiments, this experiment utilized single precision due to memory constraints. Limited reset initialization yielded the best results for initializing the vanilla GD line search. We set the learning rates for all other methods by tuning over a grid. We give the search grid as well as the selected learning rate in Table 2.

Method	Grid	Learning Rate
Fixed	$\{10^0, 10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-1}
Adam	$\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$	10^{-3}
HB	$\{10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-2}
NES	$\{10^{-1}, 10^{-2}, 10^{-3}\}$	10^{-2}

Table 2: Learning rate tuning for ResNet18 model on Imagenette.

Additional Results We now collect the additional results from our ResNet18 experiments. In Figure 15 we give the results from Figure 4 with PoNo included. Again, our method outperforms the best iterates of the PoNo line search method. In Figure 16 we compare between scalings and see that,

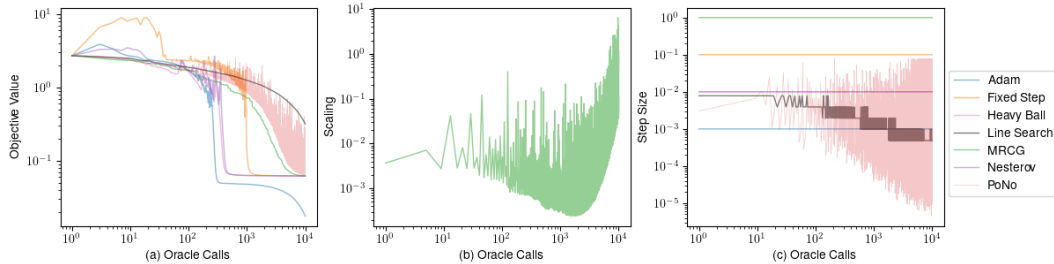


Figure 15: ResNet18 on Imagenette (Figure 4 with PoNo line search included.) (a) Objective value. (b) Scaling utilized by the MRCG method. (c) Step size. Crosses indicate iterations where negative curvature is detected.

similar to the previous two experiments, alternating MRCG scaling performs significantly better than the others.

In Figure 17 and Figure 18 we consider the CG and MR scalings in particular. Inspecting panel (b) of both figures and comparing with Figure 4, we see that the alternating MRCG scaling is capable

⁶Available here.

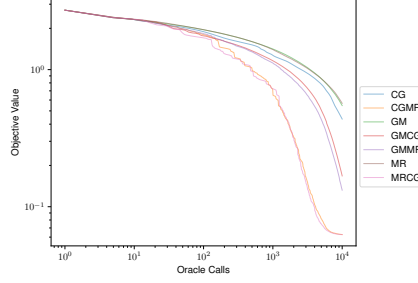


Figure 16: Comparison of scaling method performance on for the ResNet18 model on the Imagenette dataset

of producing significantly larger scaling values, which could explain the faster convergence of the alternating scaling. In panel (a) of Figure 18 we plot the gradient norm of the objective, we see that, similar to the MLP and logistic regression example, the MR scaling produces monotonic decrease in the gradient norm with the unit step size. This result is consistent with Theorem 3.9.

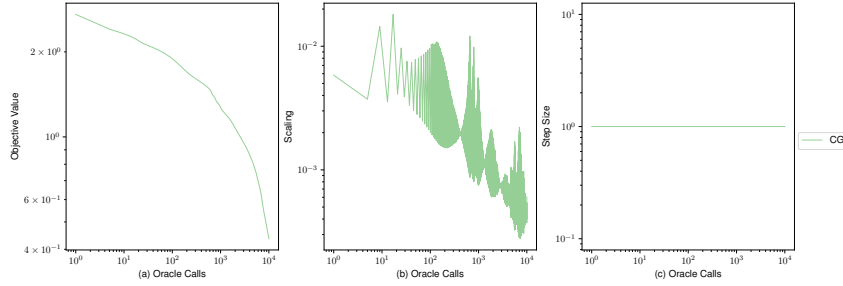


Figure 17: Performance of the CG scaling for the ResNet18 model on Imagenette dataset. (a) The objective function. (b) The scaling selected by the CG method.

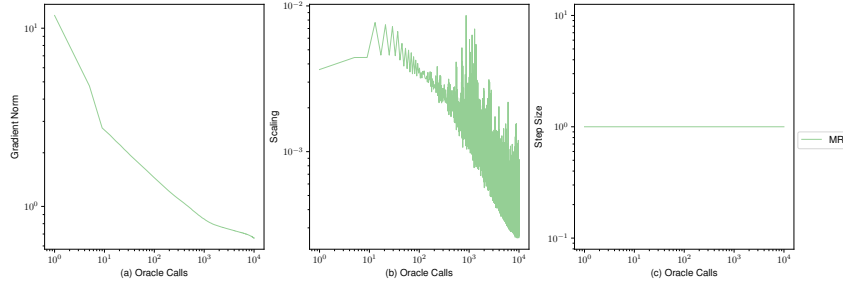


Figure 18: Performance of the MR scaling for the ResNet18 model on Imagenette dataset. (a) The objective function. (b) The scaling selected by the MR method.

In Figure 19 we plot the results from Figure 4 against wall-clock time. We see that the performance gap between wall clock and oracle calls is slightly worse relative to what was observed Figures 9 and 14. This could be due to optimizations which are unavailable for second derivative information associated with more sophisticated convolutional architectures like ResNet.

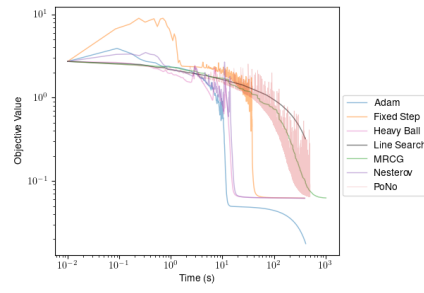


Figure 19: Wall-clock time results for the ResNet18 model on Imagenette dataset. 10^{-2} has been added to all times for plotting purposes.